Preview Report		Experimental Record		Analysis & Discussion		**Total**	

Grade & Major:	Physics, 2022	Group number:	Group 2
Student name:	**杨舒云**	Student number:	**22344020**
Experiment time:	2024/12/20	Teacher's Signature:	

APL1-2 材料真空兼容性测试和等离子特性研究

Vacuum Compatibility Testing of Materials and Plasma Characteristics Study

【Precautions】

1. The lab report consists of three parts:

   (1) **Prview Report**: Carefully study the experimental manual before class to understand the experimental principles; familiarize yourself with the instruments, equipment, and tools needed for the experiment, and their usage; complete the pre-lab thought questions; understand the physical quantities to be measured during the experiment, and prepare the experimental record forms in advance as required (you may refer to the experiment report template and print it if needed).

   (2) **Experimental Records**: Meticulously and objectively record the experimental conditions, phenomena observed during the experiment, and data collected. Experimental records should be written in ballpoint pen or fountain pen and signed (**Records written in pencil are considered invalid**). **Keep original records, including any errors and deletions; if a correction is necessary due to an error, it must be made according to the standard procedure.** (Records should not be entered into a computer and printed, but handwritten notes can be scanned and printed); before leaving, have the experimental teacher check and sign the records.

   (3) **Data Processing and Analysis**: Process the raw experimental data (except for experiments that focus on learning the use of instruments), analyze the reliability and reasonableness of the data; present the data and results in a standardized manner (charts and tables), including numbering and referencing the data, charts, and tables sequentially; analyze the physical phenomena (including answering the experimental thought questions, writing out the thought process, and citing data as needed according to standards); finally, draw a conclusion.

The experiment report combines the preparation report, experimental records, and data processing and analysis, along with this cover page.

2. Submit the **experiment report** within one week after completing each experiment (under special circumstances, no later than two weeks).

【Safety】

1. Before operation, please check whether the vacuum chamber is sealed. Ensure that the high voltage power supply switch, molecular pump power switch, and emergency button are turned off.

2. Pay attention to the safety of using the high voltage power supply. (The high voltage power supply is controlled by the vacuum gauge. Before the experiment, confirm that the vacuum gauge is powered on. Do not plug or unplug the high voltage output interface on the rear panel of the high voltage power supply while it is powered on, and avoid touching the rear-side electrical control section. Before the experiment, check the adjustment knob of the high voltage power supply and ensure it is set to zero. During the experiment, do not touch the rear panel of the high voltage power supply or any internal structures.)

3. If the experiment requires the use of a molecular pump, the mechanical pump must first reduce the vacuum pressure below 10 Pa before turning on the molecular pump power supply.

4. If a quadrupole mass spectrometer is used in the experiment, ensure that the vacuum pressure is below 5.0×10^{-2} Pa before turning it on.

【Special Note】

- Special thanks to **Huanyu Shi**, a senior from the Class of 2019, for providing the LaTeX orginal template for this experiment report, which was adjusted by **Shuyun Yang**.
- **Due to the absence of an experiment number in the original template, a self-named number has been added for ease of organization on the computer.**
- Additionally, **this experiment report is being improved towards full English expression, so there may be instances of mixed Chinese and English during this transition period. We appreciate your understanding!**
- 补充说明：预习报告由于提前预习了，因此全英文书写；实验报告主要内容（实验过程、数据记录和数据分析）由中文书写。

OUTLINE

APL1-2 Vacuum Compatibility Testing of Materials and Plasma Characteristics Study
Preview Report

1.1 Purpose

1. Learn the basic knowledge and techniques of vacuum technology, and master the methods for achieving and measuring vacuum.
2. Verify Paschen's Law through vacuum gas discharge experiments and understand the basic physical processes of gas discharge.
3. Use a fiber optic spectrometer to study the spectral characteristics of vacuum gas discharge plasma, obtain basic plasma parameters, and understand the fundamental concepts of plasma physics.
4. Understand the working principle of the quadrupole mass spectrometer, and use it for vacuum system leak detection and gas composition analysis.

1.2 Instruments & Equipment

Number	Name	Quantity	Main parameters (model, measurement range, measurement accuracy, etc.)
1	上海宜准公司 VQP01 真空平台	1	针对帕邢实验、四极质谱实验等实验项目而设计的一台综合实验装置，该装置由真空放电腔体、机械泵、分子泵、高压电源、四极质谱仪、真空计以及击穿电压测量系统等装置构成。

1.3 Principle

1. Vacuum and Related Concepts

(1) Vacuum

In physics, vacuum typically refers to a state of space that is devoid of matter, particularly lacking gas molecules. In an ideal scenario, vacuum is described as a space entirely free of any matter, but in reality, a perfect vacuum does not exist, as there will always be some residual gas molecules, cosmic rays, or dark matter. Pascal's experiments were the first to demonstrate the existence of vacuum, showing that most air could be evacuated from a sealed container to create an approximate vacuum environment. In quantum field theory, even in a complete vacuum, quantum fluctuations exist. These fluctuations indicate that vacuum is not entirely empty but is filled with the creation and annihilation of virtual particles.

The definition of vacuum varies slightly across different technical and scientific fields:

Physical Definition: From a macroscopic perspective, vacuum refers to a space devoid of matter. From a microscopic perspective, it is a state where the pressure is below a certain threshold (usually far below atmospheric pressure).

Engineering Definition: In engineering applications, vacuum often refers to any environment with pressure lower than atmospheric pressure, used in various industrial and research activities such as vacuum packaging and electron tube manufacturing.

Depending on the pressure, vacuum can be classified into several types:

- Low vacuum: Pressure slightly below atmospheric pressure.
- Medium vacuum: Pressure further reduced, generally between a few pascals and a few millipascals.
- High vacuum: Pressure even lower, typically at the micropascal level.
- Ultra-high vacuum: Extremely low pressure, reaching the nanopascal level.

(2) Mean Free Path

The mean free path is the average distance a particle travels between two successive collisions. Define $\bar{Z}$ as the average number of collisions per unit time for a single particle, and $\bar{v}$ as the average displacement per unit time. The mean free path can then be defined as $\bar{\lambda} = \frac{\bar{v}}{\bar{Z}}$, making it crucial to understand $\bar{Z}$. Here, the concept of collision cross-section is introduced. Let the radii of the two particles be r_A and r_B, then the collision cross-section between them is $\sigma = \pi(r_A + r_B)^2$. Now, considering a single particle as the reference frame:

$$\bar{Z} = \frac{\Delta N}{\Delta t} = \frac{n\sigma \bar{v_{12}}\Delta t}{\Delta t} = \sqrt{2}n\sigma\bar{v}$$

The above equation utilizes the relationship for the velocity statistics of identical particles, $\bar{v_{12}} = \sqrt{2}\bar{v}$.

In general, the mean free path is:

$$\bar{\lambda} = \frac{\bar{v}}{\bar{Z}} = \frac{1}{\sqrt{2}n\sigma} = \frac{kT}{\sqrt{2}\sigma p}$$

The higher the gas density, the shorter the distance between molecules, leading to more frequent collisions, thereby shortening the mean free path. If the molecular diameter is large or the shape is complex, the collision cross-section increases, resulting in a higher collision frequency and a reduced mean free path. As the temperature increases, the average molecular velocity $\bar{v}$ increases, which, although not directly affecting λ, may influence the density n and collision cross-section σ.

The mean free path is a key parameter in describing gas viscosity, thermal conductivity, and diffusivity. For example, the coefficient of viscosity for a gas can be expressed as $\eta = \frac{1}{3}\rho\bar{v}\lambda$. For a rarefied gas, if the container dimension is L, then $\bar{\lambda} \approx L$. In a high vacuum environment, due to the extremely low gas density, the mean free path of molecules can be larger than the dimensions of the container, meaning gas molecules can move without collisions within the container. This is important for vacuum system design and leak detection. In plasma, the mean free path of electrons and ions significantly influences plasma conductivity, reaction rates, and other properties.

(3) Transport Processes

i. For viscosity, start with the Newtonian viscosity equation:

$$f = \frac{dP}{dt} = -\eta \frac{du}{dz} A \Rightarrow J_p = \frac{dP}{Adt} = -\eta \frac{du}{dz} \Rightarrow J_p = -\eta \partial_z u$$

J_p is the tangential momentum flux density, representing the momentum transferred between different layers per unit time and unit area.

Microscopic mechanism: Fluid molecules travel between layers due to thermal motion, with thermal motion causing isotropic transfer rates, but the uneven exchange of momentum gradually reduces the momentum difference between layers.

ii. For diffusion, distinguish between self-diffusion (small overall displacement), mutual diffusion (significant overall displacement), and free diffusion (unobstructed molecular diffusion into vacuum). Diffusion is described by Fick's law:

$$J_N = -D \frac{dn}{dz} \Rightarrow J_N = -D \nabla n$$

J_N is the diffusive flux of particles, and D is the diffusion coefficient.

Microscopic mechanism: When the number density of molecules is uneven and there is no force to maintain this unevenness, a net exchange of molecular mass occurs between layers.

iii. Thermal conduction is described by Fourier's law, where J_T is the heat flux density:

$$J_T = -\kappa \frac{dT}{dz} \Rightarrow J_T = -\kappa \nabla T$$

Microscopic mechanism: For gases, thermal motion leads to the exchange of molecules with different energies; for solids and liquids, energy transfer occurs mainly through the collision of vibrating molecules.

(4) Molecular Flow (Regime): When gas pressure decreases (high vacuum), the mean free path of molecules exceeds the container dimensions, and molecules do not interact with each other but only with the container walls.

(5) Adsorption and Desorption of Gases on Solid Surfaces

► Adsorption

i. Adsorption refers to the process in which gas or liquid molecules interact with and adhere to a solid surface. Depending on the nature of the interaction forces involved in the adsorption process, adsorption can be classified into two main types—physisorption, which is more likely to occur at low temperatures due to weaker molecular thermal motion, driven mainly by van der Waals forces (weak intermolecular forces), with lower adsorption energy, typically reversible, commonly observed in the adsorption of gases on porous materials like activated carbon; chemisorption, which usually requires higher temperatures to overcome activation energy and form chemical bonds, involving covalent or ionic bonds between molecules and surface atoms, with higher adsorption energy, typically irreversible, often seen in the adsorption of reactant molecules on catalyst surfaces.

ii. The Langmuir adsorption model, proposed by Langmuir, is suitable for describing monolayer adsorption. It assumes that the adsorption surface is uniform, adsorption sites are equivalent, and adsorption is localized, meaning each adsorption site can accommodate only one adsorbed molecule. The relationship between the amount of adsorption θ and gas partial pressure P can be expressed as:

$$\theta = \frac{KP}{1+KP}$$

where K is the Langmuir adsorption constant, reflecting the strength of adsorption.

iii. BET theory extends Langmuir's theory and is used to describe multilayer adsorption, particularly in physisorption. The BET isotherm equation is:

$$\frac{1}{v[(P_0/P)-1]} = \frac{c-1}{v_m c} \cdot \frac{P}{P_0} + \frac{1}{v_m c}$$

where: v is the volume of adsorbed gas; v_m is the maximum adsorption volume for monolayer coverage; P_0 is the saturation vapor pressure; c is a constant related to the heat of adsorption.

iv. Additionally, using the grand canonical ensemble, we can also derive:

$$\bar{N} = -\frac{\partial}{\partial \alpha} \ln \Xi = \frac{N_0}{1+e^{-\beta(\mu+\varepsilon_0)}} \quad \Rightarrow \quad \theta = \frac{1}{1+\frac{kT}{p}\left(\frac{2\pi mkT}{h^2}\right)^{3/2} e^{-\frac{\varepsilon_0}{kT}}}$$

The above equation shows that θ increases with the rise of p and decreases with the rise of T.

- Desorption

  Desorption refers to the process in which gas molecules adsorbed on a solid surface gain energy and detach from the surface. Desorption can be achieved through the following methods:

  Thermal Desorption: By heating the surface, enough energy is provided for the adsorbed molecules to overcome the adsorption energy and detach from the surface. This method is commonly used to study adsorption energy and adsorption site distribution.

  Pressure Desorption: By lowering the partial pressure of the gas, the interaction forces between the adsorbed molecules and the surface are reduced, making desorption easier. This method is often used in the regeneration process of porous materials, such as the regeneration of activated carbon.

Photodesorption: Using high-energy photons (such as ultraviolet light) to irradiate the adsorbed surface, the energy of the photons is absorbed by the adsorbed molecules, causing desorption. This process is common in photocatalysis and photochemical reactions.
Electric or Magnetic Field-Assisted Desorption: By applying an electric or magnetic field, the interaction between the adsorbed molecules and the surface is altered, leading to desorption. This method is used in certain specific studies of electrochemical or magnetic materials.

(6) Evaporation of Liquid Surface Molecules and Sublimation of Solid Surface Molecules

- Evaporation is the process where liquid molecules escape from the liquid surface and enter the gas phase. Evaporation typically occurs at temperatures below the boiling point of the liquid and only at the liquid surface.
  Due to thermal motion, liquid molecules constantly collide within the liquid. The movement speeds of these molecules vary, and some molecules have higher kinetic energy. Surface molecules experience less intermolecular force as they lack the confinement from molecules outside the liquid. If surface molecules possess sufficient kinetic energy to overcome the surface tension, they can escape from the liquid surface and enter the gas phase, forming evaporation. In a closed system, the gas molecules escaping during evaporation will gradually accumulate until equilibrium is reached, at which point the rate of evaporation equals the rate of condensation, and the number of molecules between the gas and liquid phases remains constant.
  The higher the temperature, the more intense the molecular motion, increasing the number of molecules with sufficient kinetic energy to escape, hence increasing the evaporation rate. The larger the liquid surface area, the more molecules participate in evaporation, so the evaporation rate is proportional to the surface area. In a low-pressure environment, the reflux of gas molecules decreases, increasing the evaporation rate. The higher the humidity, the more water vapor molecules present in the air, inhibiting the evaporation of more liquid molecules. Therefore, the higher the humidity, the lower the evaporation rate.
- Sublimation is the process where a solid directly transitions to a gas, bypassing the liquid phase. This process typically occurs under specific conditions, such as low temperature or low pressure, or in solids with relatively weak intermolecular forces.
  In solids, molecules or atoms are fixed in a lattice structure by strong intermolecular forces (such as van der Waals forces, hydrogen bonds, or ionic bonds). However, a small number of molecules may have higher kinetic energy. Under specific conditions, surface molecules of a solid can gain sufficient kinetic energy to overcome intermolecular forces and directly escape into the gas phase. These molecules bypass the liquid phase and directly change from solid to gas. Similar to evaporation, in a closed system, sublimation will reach dynamic equilibrium with deposition (the direct transition of gas to solid), where the rate of sublimation equals the rate of deposition.
  The phenomenon of sublimation becomes significantly enhanced with rising temperature

because higher temperatures provide more kinetic energy for molecules to escape from the solid surface. At lower pressures, solids are more prone to sublimation due to the reduced binding force between molecules under low pressure. The sublimation capability of different solids depends on their internal intermolecular forces. For example, dry ice (solid carbon dioxide) and iodine have strong sublimation capabilities, as the intermolecular forces in these substances are relatively weak under normal temperature and pressure.

(7) Vacuum Generation

i. Mechanical Pumps

Mechanical pumps are commonly used devices to create rough and medium vacuum by removing gases from the system through mechanical motion.

- Rotary Vane Pump:
  - Working Principle: The rotary vane pump uses a rotor installed eccentrically in the pump body. The vanes on the rotor capture and compress the gas, eventually discharging it.
  - Applications: Suitable for creating rough vacuum and is often used as a forepump in many high vacuum systems.
- Oil-Sealed Rotary Pump:
  - Working Principle: The oil-sealed rotary pump uses oil to seal and lubricate internal mechanical components, improving the pump's efficiency and longevity.
  - Applications: Widely used in laboratories, industrial manufacturing, and scientific research for rough vacuum generation.
- Dry Pump:
  - Working Principle: The dry pump does not use oil as a sealant or lubricant. It compresses and discharges gas through oil-free mechanical methods.
  - Applications: Suitable for environments requiring oil-free vacuum, such as semiconductor manufacturing.

ii. Molecular Pumps

Molecular pumps are used to achieve high and ultra-high vacuum by relying on the interaction between gas molecules and a high-speed rotating surface to drive gas molecules out of the system.

- Turbo Molecular Pump (TMP):
  - Working Principle: The turbo molecular pump uses high-speed rotating turbine blades to compress and discharge gas molecules step by step. The design of the turbine blades causes gas molecules to be gradually pushed toward the exhaust port through multiple collisions.
  - Applications: Used for high and ultra-high vacuum generation, widely applied in electron microscopes, mass spectrometers, and semiconductor production equipment.

- Diffusion Pump:
  - Working Principle: The diffusion pump uses high-temperature oil vapor to form a high-speed jet stream. Gas molecules are captured by the vapor stream and expelled from the system. The diffusion pump has no moving parts, resulting in high reliability and long service life.
  - Applications: Suitable for high vacuum environments and commonly used as the main pump in electron beam welding and vacuum coating equipment.

iii. Sorption Pumps

Sorption pumps use adsorbent materials (such as activated carbon or molecular sieves) at low temperatures to capture and adsorb gas molecules, thereby achieving vacuum.

- Cryopump:
  - Working Principle: The cryopump lowers the gas to extremely low temperatures, causing it to condense and adsorb onto a cold surface, effectively reducing the gas pressure in the system.
  - Applications: Commonly used in applications requiring a clean vacuum environment, such as semiconductor manufacturing and superconducting research.
- Activated Charcoal Pump:
  - Working Principle: Activated charcoal has a large number of micropores, which can effectively adsorb gas molecules at low temperatures, thereby reducing the pressure.
  - Applications: Typically used in conjunction with other pumps, especially in the process of achieving ultra-high vacuum.

iv. Ion Pumps

Ion pumps achieve gas removal by ionizing gas molecules and accelerating them to a solid surface. They are commonly used to obtain and maintain ultra-high vacuum.

- Sputter Ion Pump:
  - Working Principle: Gas molecules are ionized in a strong electric field, and the resulting ions are accelerated to a metal electrode and captured or combined through the sputtering effect.
  - Applications: Widely used in high-energy physics, particle accelerators, and surface science research, where long-term maintenance of ultra-high vacuum is required.
- Getter Ion Pump:
  - Working Principle: Uses getter materials (such as titanium) to capture and adsorb ionized gas molecules, further enhancing vacuum through the ionization effect of the ion pump.
  - Applications: Used in applications requiring long-term maintenance of ultra-high vacuum, such as space simulation experiments and precision measurement instruments.

v. Combined Vacuum Systems

To achieve a gradual transition from rough vacuum to ultra-high vacuum, multiple vacuum pumps are often used in combination. Such combined systems typically include:

- Backing Pump: Such as rotary vane pumps or oil-sealed pumps, used for initially lowering the pressure in the system.
- Main Pump: Such as turbo molecular pumps or diffusion pumps, used to further reduce the pressure to achieve high vacuum.
- Auxiliary Pump: Such as ion pumps or cryopumps, used to maintain ultra-high vacuum or remove residual gases.

2. Gas Discharge and Plasma

- Gas Discharge

  Gas discharge refers to the ionization process of a gas under an applied electric field, which causes the gas to transition from an insulator to a conductor, resulting in the generation of an electric current through the gas. Gas discharge is widely used in everyday life and industry, such as in neon lights, fluorescent lamps, discharge tubes, and plasma technologies.

  (1) Basic Process of Gas Discharge

  A gas is originally an electrical insulator, but under the influence of an applied electric field, gas molecules may become ionized into free electrons and positive ions. This process usually requires collisions, particularly when the electric field is strong enough for electrons to gain sufficient energy to ionize gas molecules through collisions.

  The initially produced free electrons are accelerated in the electric field and may collide with other gas molecules, further producing more electrons and ions. This avalanche effect rapidly enhances the gas's conductivity.

  When the number of free electrons in the gas reaches a certain level, the gas's conductivity increases sharply, and the current suddenly increases, forming a gas discharge. During this process, the path of the current through the gas may emit light (such as the glow discharge in neon lights) and may be accompanied by other phenomena, such as sound or arcs.

  (2) Paschen's Law

  Breakdown voltage refers to the minimum voltage required under specific conditions to cause the gas to undergo breakdown and transition to a conductive state. The breakdown voltage is closely related to the electrode gap (the distance between the electrodes) and gas pressure. To understand this relationship, we must delve into Paschen's Law. Paschen's Law, proposed by the German physicist Friedrich Paschen in 1889 through experiments, describes the relationship between the breakdown voltage V_b and the product of electrode gap d and gas pressure p when gas breakdown occurs between two electrodes.

  The mathematical expression of Paschen's Law is:

$$V_b = \frac{B \cdot pd}{\ln(A \cdot pd) - \ln[\ln\left(1 + \frac{1}{\gamma}\right)]}$$

where V_b is the breakdown voltage; p is the gas pressure; d is the distance between the electrodes; A and B are constants related to the properties of the gas; γ is the secondary electron emission coefficient, representing the efficiency of secondary electron emission caused by a positive ion striking the cathode.

Paschen's Law indicates that the breakdown voltage V_b does not simply increase with the electrode gap or gas pressure but has a more complex relationship with their product pd. For a specific gas and electrode material, there exists a particular pd value where the breakdown voltage V_b reaches its minimum. This implies that in some cases, by adjusting the electrode gap and gas pressure, the breakdown voltage can be minimized, which is particularly important in the design of discharge devices. When pd increases to a large value, the breakdown voltage V_b also increases. This is because at larger gaps or higher pressures, electrons require more energy (higher voltage) to gain enough energy within a single mean free path to ionize gas molecules. When pd decreases to a very small value, the breakdown voltage V_b rises sharply. This is because, at low pressures, the gas molecule density is low, and collisions between electrons are less frequent, leading to the need for higher voltage to ionize the gas and cause breakdown.

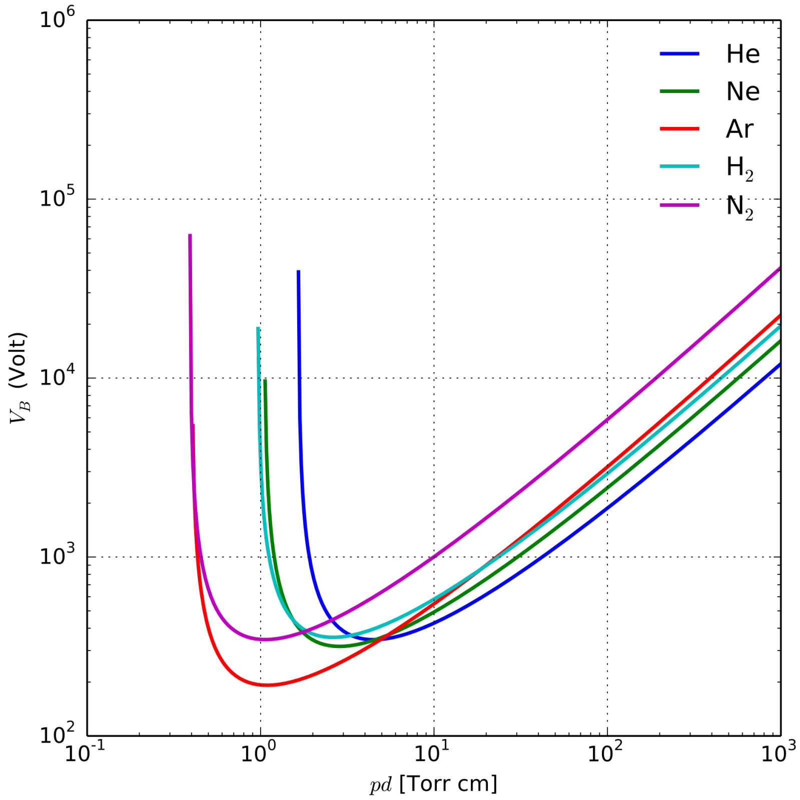

Figure 1: Paschen curves obtained for helium, neon, argon, hydrogen and nitrogen, using the expression for the breakdown voltage as a function of the parameters A,B that interpolate the first Townsend coefficient (From reference [1]).

The Paschen curve is a graphical representation of Paschen's Law, showing the relationship between the breakdown voltage V_b and pd. The Paschen curve typically exhibits a "U" shape, indicating that at a certain pd value, the breakdown voltage reaches its minimum. When pd is very small, the breakdown voltage is high, and it is difficult for the gas to break down. This is because at low gas pressure, the molecule density is low, and it is hard for electrons to gain enough energy to cause ionization. When pd is very large, the breakdown voltage is also high, as the large electrode gap or high pressure causes electrons to lose more energy before reaching the other electrode, necessitating higher voltage to break down the gas. The minimum point at the bottom of the curve corresponds to the minimum breakdown voltage, and the pd value at this point is referred to as the "optimal condition."

- Plasma

Plasma consists of free electrons, ions, charged particles, and neutral atoms or molecules. Since the particles in plasma are charged, it can respond to electromagnetic fields and exhibits unique physical properties. Gas discharge is a common method of generating plasma.

(1) Basic Parameters of Plasma

Electron Density (n_e): Refers to the number of free electrons per unit volume, usually expressed as the number of electrons per cubic centimeter. Electron density determines the conductivity and electromagnetic response of the plasma.

Ion Density (n_i): Refers to the number of ions per unit volume. For most plasmas, the electron density and ion density are equal, reflecting the electrical neutrality of the plasma.

Electron Temperature (T_e): Electron temperature refers to the average kinetic energy of electrons in the plasma, typically expressed in Kelvin (K) or electron volts (eV). Electron temperature determines the degree of ionization and chemical reaction rates in the plasma.

Ion Temperature (T_i): Similar to electron temperature, ion temperature represents the average kinetic energy of ions. Ion temperature is usually lower than electron temperature, especially in cold plasmas.

Plasma Frequency (ω_p): Plasma frequency is the frequency at which electrons in the plasma oscillate collectively. It depends on electron density and is defined as:

$$\omega_p = \sqrt{\frac{n_e e^2}{\epsilon_0 m_e}}$$

where n_e is the electron density, e is the electron charge, m_e is the electron mass, and ϵ_0 is the vacuum permittivity.

Debye Length (λ_D): Debye length is the spatial scale over which potential changes significantly in the plasma, defined as:

$$\lambda_D = \sqrt{\frac{\epsilon_0 k_B T_e}{n_e e^2}}$$

where k_B is the Boltzmann constant, and T_e is the electron temperature. The Debye length reflects the plasma's ability to shield external electric fields.

Magnetic Field (B): In magnetized plasma, the interaction between an external magnetic field and the plasma significantly influences its behavior. For example, in fusion reactors, magnetic fields are used to confine plasma to achieve stable fusion reactions.

(2) Plasma Spectroscopy

Plasma spectroscopy is a method of studying the properties of plasma by analyzing the electromagnetic radiation spectrum it emits. Plasma spectroscopy includes atomic spectra, ionic spectra, molecular spectra, and continuous spectra, reflecting the physical state and chemical composition of the plasma. When atoms or ions in the plasma are excited and then return to lower energy levels, photons are emitted. Emission spectroscopy analyzes the energy of photons from transitions between different energy levels in the plasma, revealing its physical properties. When light from an external source passes through the plasma, specific wavelengths of light may be absorbed by particles in the plasma, reducing the intensity of light at those wavelengths. By measuring these absorption spectra, the density and temperature of particles in the plasma can be inferred. When light interacts with particles in the plasma, the direction and wavelength of the light may change, producing scattering spectra. This method is mainly used to study the microscopic structure and particle motion in plasma. Spectra caused by the acceleration or deceleration of free electrons in electric or magnetic fields (e.g., bremsstrahlung or free-free transitions) typically present as a continuous spectral distribution.

By detecting the characteristic spectral lines of specific elements or compounds, the chemical composition of the plasma can be determined. By analyzing the width, intensity, and shift of spectral lines, the electron temperature and ion temperature of the plasma can be inferred. For example, the broadening of spectral lines can reflect the thermal motion of electrons or ions, which corresponds to the electron temperature or ion temperature. Electron temperature is usually determined by analyzing the continuous spectrum (such as bremsstrahlung) or the intensity distribution of emission lines. By analyzing the intensity distribution of spectral lines and the relative intensity between lines, the electron density and ion density in the plasma can be estimated. For example, the broadening of spectral lines due to the Stark effect is proportional to the electron density, thus it can be used to measure electron density. The broadening of spectral lines due to the Doppler effect can be used to analyze the velocity distribution of particles in the plasma, thereby studying the kinetic properties of the plasma. For example, in plasmas under strong magnetic fields, the Doppler effect can reveal the motion of particles in the direction of the magnetic field.

3. Quadrupole Mass Spectrometer

The quadrupole mass spectrometer is a widely used mass spectrometry instrument that separates and detects ions with different mass-to-charge ratios (m/z) by generating a specific quadrupole electric field. It is extensively applied in chemical analysis, environmental monitoring, pharmaceutical research, and other fields. The core of the quadrupole mass spectrometer lies in controlling the motion of ions in the quadrupole electric field to achieve ion separation and mass analysis.

(1) The quadrupole mass spectrometer mainly consists of the following components:

- Ion Source: Used to ionize sample molecules, generating charged ions.
- Quadrupole Rod System: Composed of four parallel conductive rods, forming a quadrupole electric field by applying direct current (DC) and radio frequency (RF) voltages.
- Detector: Used to detect ions passing through the quadrupole electric field and record their signals.

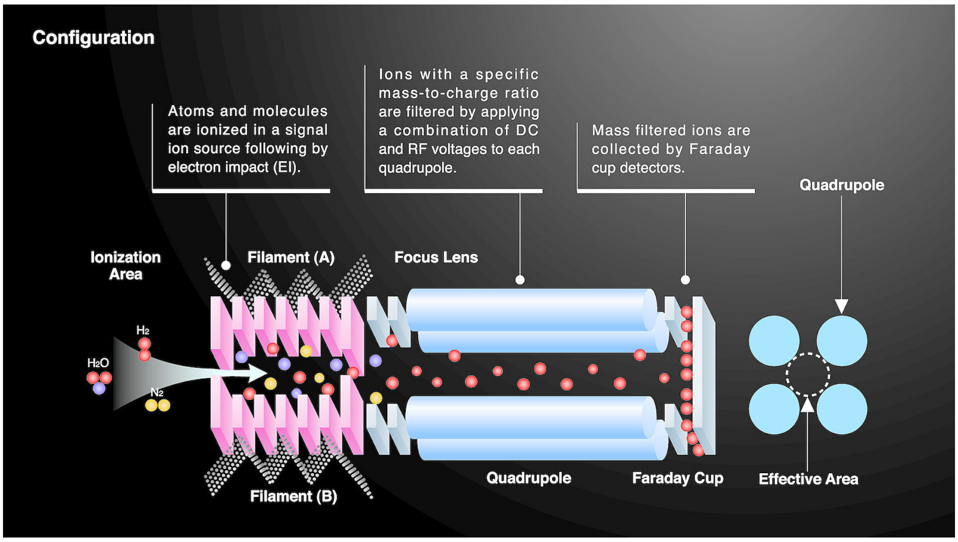

Figure 2: Schematic diagram of the principle of a quadrupole mass spectrometer (From web).

(2) Characteristics of the Quadrupole Electric Field

In the quadrupole mass spectrometer, four parallel conductive rods (typically metal bars) are arranged in pairs, with different voltages applied to generate a quadrupole electric field. The characteristic of this quadrupole electric field is that the electric field strength varies with position in space and has specific symmetry.

The voltage applied to the quadrupole rod system consists of a direct current voltage U and a radio frequency voltage V. For a pair of opposite quadrupole rods, the voltage can be expressed as:

$$\phi(x, y, t) = \frac{U + V\cos(\omega t)}{r_0^2}\left(x^2 - y^2\right)$$

where $\phi(x, y, t)$ is the electric potential; U is the direct current voltage; V is the peak value of the radio frequency voltage; ω is the angular frequency of the radio frequency voltage; r_0 is the effective radius inside the quadrupole rods; x and y are the coordinates in the quadrupole rod plane.

The electric field generated by this potential distribution is time-dependent and has no field component in the z axis direction (along the axis of the quadrupole rods), with field components only in the x and y directions.

(3) Ion Motion Equations in the Quadrupole Electric Field

To deeply understand the working principle of the quadrupole mass spectrometer, one must study the motion of ions in the quadrupole electric field. This problem can be described by solving the Mathieu equation in classical mechanics.

Consider an ion with mass m and charge q, subjected to a force $F = qE$ in the quadrupole electric field, where E is the electric field strength. The ion's motion in the x and y directions can be described by the following differential equations:

$$\frac{d^2x}{dt^2} + \frac{q}{m}\frac{U + V\cos(\omega t)}{r_0^2}x = 0$$

$$\frac{d^2y}{dt^2} - \frac{q}{m}\frac{U + V\cos(\omega t)}{r_0^2}y = 0$$

By introducing the dimensionless parameters a_x and q_x, the above differential equations can be normalized to the standard form of the Mathieu equation:

$$\frac{d^2u}{d\xi^2} + [a - 2q\cos(2\xi)]\,u = 0$$

where u can represent either x or y; $\xi = \frac{\omega t}{2}$ is the normalized time; $a_x = \frac{4qU}{mr_0^2\omega^2}$ and $q_x = \frac{2qV}{mr_0^2\omega^2}$ are dimensionless parameters.

The solutions of the Mathieu equation can be classified into two categories: stable solutions and unstable solutions. Stable solutions correspond to stable ion motion in the quadrupole electric field, while unstable solutions correspond to ion divergence or ejection from the quadrupole rod system. By analyzing the solutions of the Mathieu equation, one can determine the stability regions for ion motion under specific a and q parameters, corresponding to ions with specific mass-to-charge ratios m/z that can be stably transmitted through the quadrupole rod system. Ions in the unstable regions will be ejected by the alternating electric field and will not reach the detector.

(4) Working Principle of the Quadrupole Mass Spectrometer

During operation, the quadrupole mass spectrometer separates ions with different mass-to-charge ratios by adjusting the direct current voltage U and the radio frequency voltage V applied to the quadrupole rods, controlling the stability region of ions. For a specific m/z, precise adjustment of the U and V ratio ensures that the ion remains in a stable motion state in the quadrupole electric field, allowing it to pass through the quadrupole rods and reach the detector. By linearly or stepwise adjusting U and V, different m/z ranges can be scanned, gradually transmitting ions with different mass-to-charge ratios. After the detector records the signals of different ions, a mass spectrum can be plotted, showing the relative abundance of different components in the sample.

The resolution of the quadrupole mass spectrometer depends on the width of the stability region through which ions pass in the quadrupole electric field. Higher resolution means more precise differentiation of ions with close masses. Additionally, the frequency and amplitude of the radio frequency voltage also significantly affect resolution and sensitivity.

1.4 Thinking Before Experiment

Reflection Question 1.1:

What are the research topics and methods in vacuum physics?

Vacuum physics is the theoretical foundation of vacuum science and technology, focusing on the theoretical study of the physical behavior of rarefied gases. It primarily employs methods from statistical physics and thermodynamics to investigate the motion of gas molecules and the interactions between gas molecules as well as between molecules and the walls of containers.

Reflection Question 1.2:

What is the definition of vacuum? What is the ideal gas pressure formula? What is the mean free path of gas molecules?

1. Definition of Vacuum: A state in a given space where the gas pressure is lower than one standard atmosphere (the state in which the density of gas molecules is lower than the density of gas molecules at standard atmospheric pressure) is referred to as a vacuum.

2. Ideal Gas Pressure Formula:

   The equation of state for an ideal gas is:

   $$pV = nRT$$

   From this equation, the pressure formula for an ideal gas, which relates the gas pressure p to the molecular number density n, can be derived as:

   $$p = nk_BT$$

   where p is the gas pressure; n is the number density of gas molecules, i.e., the number of molecules per unit volume, with the unit m^{-3}; k_B is the Boltzmann constant, valued at 1.38×10^{-23} J/K; and T is the absolute temperature of the gas.

   This formula indicates that at a given temperature, the gas pressure is directly proportional to the number density of the gas molecules.

3. Mean Free Path of Gas Molecules: See the principle overview section for details; the formula is provided below.

$$\bar{\lambda} = \frac{\bar{v}}{\bar{Z}} = \frac{1}{\sqrt{2}n\sigma} = \frac{kT}{\sqrt{2}\sigma p}$$

Reflection Question 1.3:

What are the basic processes of vacuum gas discharge? What is Paschen's Law?

- Vacuum Gas Discharge refers to the process where gas becomes ionized under the influence of an applied electric field in a vacuum or low-pressure environment, leading to the transition of the gas from an insulator to a conductor. The process of gas discharge under vacuum or low-pressure conditions can be divided into the following key stages:

  1. Initial Ionization:

  Free electrons in the gas are typically excited by external factors (such as ultraviolet radiation, cosmic rays, thermionic emission, etc.). The initial number of free electrons is very small, but they serve as the seeds for the discharge process. Under the influence of an applied electric field, these free electrons accelerate in the direction of the electric field. If the electric field strength is sufficient, electrons can gain enough kinetic energy.

  2. Collision Ionization:

  Accelerated electrons may collide with gas molecules or atoms during their motion. If the electron' s kinetic energy exceeds the ionization energy of the gas molecule, it may ionize the molecule, producing a positive ion and another free electron. The newly generated free electron also accelerates under the electric field and may further trigger new ionizations. In this way, the number of free electrons and positive ions in the gas rapidly increases, forming an electron avalanche effect.

  3. Gas Conductivity:

  As the number of free electrons and ions increases significantly, the gas begins to exhibit conductivity, forming a plasma state. At this point, the gas transitions from an electrical insulator to a conductor, allowing current to flow. Depending on the electric field strength, gas pressure, and electrode spacing, gas discharge can be categorized into different types such as glow discharge, arc discharge, and spark discharge.

  4. Equilibrium State:

  Under the continuous influence of the electric field, the processes of ionization and recombination (recombination of electrons with ions) will reach a dynamic equilibrium, resulting in a stable current, and the gas discharge enters a steady state. If the applied electric field is weakened or removed, the ionization process will cease, electrons and ions will gradually recombine, and the gas will return to its insulating state, ending the discharge.

- Paschen's Law: See the principle overview section for details; a brief summary is provided below.

  The mathematical expression of Paschen's Law is:

$$V_b = \frac{B \cdot pd}{\ln(A \cdot pd) - \ln[\ln\left(1 + \frac{1}{\gamma}\right)]}$$

  where V_b is the breakdown voltage; p is the gas pressure; d is the distance between the electrodes; A and B are constants related to the properties of the gas; γ is the secondary electron emission coefficient, i.e., the efficiency of secondary electron emission caused by a positive ion striking the cathode.

  Paschen's Law indicates that the breakdown voltage V_b does not simply increase with increasing electrode spacing or gas pressure, but rather has a more complex relationship with their product pd. For a specific gas and electrode material, there exists a particular pd value at which the breakdown voltage V_b reaches a minimum. This suggests that in certain cases, the breakdown voltage can be minimized by adjusting the electrode spacing and gas pressure, which is crucial in the design of discharge devices.

Reflection Question 1.4:

What are the characteristics of glow discharge?

1. Basic Characteristics of Glow Discharge

   Glow discharge typically occurs in low-pressure environments, with gas pressures ranging from a few hundred pascals to several hundred pascals. This is because, at low pressure, the mean free path between gas molecules is relatively long, allowing electrons to accelerate under the influence of the electric field and gain enough energy for collision ionization, thereby sustaining the discharge process. Glow discharge requires a certain voltage range to occur. The voltage must be sufficiently high for electrons to acquire enough kinetic energy to ionize gas molecules. As the voltage increases, the discharge first undergoes a dark discharge stage before entering the glow discharge stage. In glow discharge, gas molecules are ionized by collisions with free electrons or ions, forming a plasma. This plasma consists of electrons, positive ions, negative ions, and neutral particles, and can emit light during the discharge process. One of the notable features of glow discharge is the emission of visible light from the discharge region, which originates from the excitation and radiative transitions of gas molecules after collisions with electrons and ions. Different gases emit different glow colors; for example, neon emits an orange-red light, while argon emits purple light.

2. Spatial Distribution Characteristics of Glow Discharge

   The physical process of glow discharge is complex and exhibits multiple spatial regions, each with its unique physical characteristics and discharge mechanisms. A typical glow discharge can be divided into the following regions:

   The Cathode Dark Space, which is adjacent to the cathode surface. Due to the low energy of the electrons, they are unable to excite gas molecules to emit light, resulting in a dark region. The electron density is low, but the ion density is high. The electric field is strongest in this region, where positive ions accelerate in the electric field, striking the cathode and causing secondary electron emission.

The Cathode Glow, located next to the cathode dark space. In this region, high-energy electrons collide with gas molecules, exciting them to emit light, which manifests as bright glow. The electric field in this region is relatively strong, with an increased density of electrons and ions.

The Faraday Dark Space, located after the cathode glow region. Electrons lose some energy in the cathode glow region, rendering them unable to effectively excite gas molecules to emit light, resulting in a dark state in the Faraday dark space. The electric field intensity decreases, but some ionization still occurs.

The Anode Glow, near the anode region. Electrons are further accelerated near the anode, exciting gas molecules to emit light, forming the anode glow. The electron density is high, and the electric field strength near the anode increases again.

The Anode Dark Space, between the anode glow and the anode. The electric field intensity in this region further weakens, and the electron energy is insufficient to excite gas molecules to emit light, resulting in a dark region. The ion flux density is relatively high, and the recombination effect of electrons and ions near the anode is significant.

3. Current-Voltage Characteristics of Glow Discharge

Glow discharge has a specific current-voltage characteristic curve (I-V curve), which reflects the relationship between voltage and current during the discharge process.

At lower voltages, the current is very small and unstable, with the discharge process mainly involving the migration of a small number of electrons and ions in the electric field, without significant gas ionization or light emission. When the voltage increases to a certain level, the discharge enters the glow discharge stage, where gas ionization intensifies, the current increases rapidly, and obvious glow phenomena appear. The discharge voltage approaches the breakdown voltage described by Paschen's law. As the voltage continues to increase, the current and the light intensity of the discharge region remain relatively stable, forming a steady-state glow discharge. At this point, the relationship between current and voltage exhibits linear growth. If the voltage continues to increase, the current will exhibit nonlinear growth, the glow region will expand, potentially leading to discharge instability or even transition to arc discharge.

Reflection Question 1.5:

What is plasma? What are the basic parameters of plasma?

For further details, please refer to the principle overview section.

Plasma consists of free electrons, ions, charged particles, and neutral atoms or molecules. Since the particles in plasma are charged, it can respond to electromagnetic fields and possesses unique physical properties. Gas discharge is a common form of plasma generation.

The basic parameters of plasma are:

1. Electron Density, n_e: Represents the number of free electrons per unit volume, usually expressed as the number of electrons per cubic centimeter. Electron density determines the conductivity and electromagnetic response capability of the plasma.

2. Ion Density, n_i: Represents the number of ions per unit volume. For most plasmas, the electron density and ion density are equal, reflecting the electrical neutrality of the plasma.

3. Electron Temperature, T_e: Refers to the average kinetic energy of electrons in the plasma, usually expressed in kelvins (K) or electron volts (eV). Electron temperature determines the degree of ionization of the plasma and the rate of chemical reactions.

4. Ion Temperature, T_i: Similar to electron temperature, ion temperature represents the average kinetic energy of ions. Ion temperature is usually lower than electron temperature, especially in cold plasmas.

5. Plasma Frequency, ω_p: The plasma frequency is the frequency at which electrons in the plasma oscillate collectively. It depends on electron density and is defined as:

$$\omega_p = \sqrt{\frac{n_e e^2}{\epsilon_0 m_e}}$$

   where n_e is the electron density, e is the electron charge, m_e is the electron mass, and ϵ_0 is the vacuum permittivity.

6. Debye Length, λ_D: The Debye length is the spatial scale over which potential changes significantly in plasma, defined as:

$$\lambda_D = \sqrt{\frac{\epsilon_0 k_B T_e}{n_e e^2}}$$

   where k_B is the Boltzmann constant and T_e is the electron temperature. The Debye length reflects the plasma's ability to shield external electric fields.

7. Magnetic Field, B: In magnetized plasmas, the interaction between the applied magnetic field and the plasma significantly influences the behavior of the plasma. For example, in nuclear fusion reactors, magnetic fields are used to confine the plasma to achieve stable fusion reactions.

Reflection Question 1.6:

What are the characteristics of the quadrupole electric field in a quadrupole mass spectrometer? How do ions move within it? What are the basic principles and processes of a quadrupole mass spectrometer?

1. Characteristics of the Quadrupole Electric Field

The quadrupole electric field is generated by two independent voltage sources: a direct current voltage U and an alternating radio frequency voltage $V\cos(\omega t)$. Four parallel metal rods are arranged in pairs to form two opposing electrode pairs. These electrode pairs are subjected to the following voltages: one pair of electrodes receives $+U+V\cos(\omega t)$, while the other pair receives $-U-V\cos(\omega t)$. Here, U is the direct current voltage, providing a constant electric field component; V is the peak value of the radio frequency voltage, producing an alternating electric field component; and ω is the angular frequency of the radio frequency voltage. In this configuration, there is no electric field component along the axis (commonly defined as the z-axis) at the center of the quadrupole rods, and the distribution of the electric field varies only in the plane perpendicular to the electrodes (i.e., the x-y plane).

The electric potential of the quadrupole field can be expressed as:

$$\phi(x, y, t) = \frac{U + V\cos(\omega t)}{r_0^2}\left(x^2 - y^2\right)$$

where $\phi(x, y, t)$ is the electric potential; x and y are the coordinates in the plane; and r_0 is the effective radius within the quadrupole rods.

From the potential, the electric field strength ($\mathbf{E}$) can be derived as:

$$E_x = -\frac{\partial \phi}{\partial x} = -\frac{2(U + V\cos(\omega t))x}{r_0^2}$$

$$E_y = -\frac{\partial \phi}{\partial y} = \frac{2(U + V\cos(\omega t))y}{r_0^2}$$

2. Ion Motion in a Quadrupole Electric Field

For an ion with mass m and charge q, the equation of motion in the quadrupole electric field can be expressed as:

$$m\frac{d^2x}{dt^2} = qE_x = -\frac{2q(U + V\cos(\omega t))x}{r_0^2}$$
$$m\frac{d^2y}{dt^2} = qE_y = \frac{2q(U + V\cos(\omega t))y}{r_0^2}$$

By appropriate variable substitution, the above equations can be simplified to the standard form of the Mathieu equation:

$$\frac{d^2u}{d\xi^2} + (a - 2q\cos(2\xi))u = 0$$

where u can represent either x or y; $\xi = \frac{\omega t}{2}$ is the normalized time; $a_x = \frac{4qU}{mr_0^2\omega^2}$ and $q_x = \frac{2qV}{mr_0^2\omega^2}$ are dimensionless parameters.

The solutions to the Mathieu equation can be classified into two categories: stable solutions and unstable solutions. Stable solutions correspond to the stable motion of ions in the quadrupole electric field, while unstable solutions correspond to the divergence or ejection of ions from the quadrupole system. By analyzing the solutions of the Mathieu equation, one can determine the regions of stable ion motion under specific parameters a and q, which correspond to ions with equivalent mass-to-charge ratios m/z that can be stably transmitted through the quadrupole system. Ions in unstable regions will be expelled from the quadrupole due to the alternating electric field and will not reach the detector.

3. Basic Principles and Process of Quadrupole Mass Spectrometry

Ionization: First, the sample is ionized by an ion source (e.g., electron impact source, chemical ionization source, etc.), generating positively or negatively charged ions.

Selective Electric Field of the Quadrupole: By applying specific U and V voltages, the characteristics of the quadrupole electric field are adjusted so that ions with a specific mass-to-charge ratio (m/z) can stably travel through the quadrupole, while ions with other mass-to-charge ratios are excluded due to unstable motion.

Mass Filtering: The quadrupole mass spectrometer can sequentially allow ions with different mass-to-charge ratios to pass stably through the quadrupole by changing the ratio of U and V. This method of scanning different mass-to-charge ratios by varying the voltage is known as the mass scanning mode.

Ion Detection: Ions passing through the mass analyzer (i.e., the quadrupole system) enter a detector (e.g., an electron multiplier), which converts the ion signals into electrical signals.

Signal Processing and Data Analysis: The electrical signals are amplified and processed to generate a mass spectrum, showing the mass-to-charge ratios and relative abundances of the components in the sample.

Reflection Question 1.7:

Why must the mechanical pump first reduce the vacuum pressure below 10 Pa before turning on the molecular pump power supply?

- Molecular Flow Conditions: The design of molecular pumps is intended to operate in molecular flow conditions, meaning the mean free path of gas molecules (i.e., the average distance between molecules) should be much greater than the characteristic size of the pump. At high pressures, the molecular spacing is smaller, and the molecular pump cannot effectively transport and expel gas molecules in a specific direction. Therefore, molecular pumps can only operate efficiently at low pressures (typically below 10 Pa).

- Gas Collisions and Pumping Efficiency: At high pressures, gas molecules frequently collide, resulting in viscous flow characteristics. In this situation, the interaction between the pump blades and gas molecules is no longer dominated by single molecule collisions with the blades but rather by molecular collisions, leading to a significant reduction in pumping efficiency. Mechanical pumps are used to reduce the pressure to a suitable level so that gas flow enters the molecular flow regime, allowing the molecular pump to function effectively.

- Operational Requirements: Molecular pumps must be started in low-pressure environments because starting a molecular pump at high pressure not only fails to effectively evacuate but may also cause damage to the internal components of the pump due to excessive gas load. High pressure increases the resistance of the turbine blades, making it difficult to start the pump and potentially causing mechanical failure. Different types of gases exhibit different flow behaviors at varying pressures. In particular, some gases may be difficult to expel through the molecular pump at high pressures, so it is necessary to pre-evacuate the system to a lower pressure using a mechanical pump before starting the molecular pump. When operating, the high-speed rotation of the pump blades generates heat. Starting the molecular pump at high pressures may cause friction between the blades and the gas, leading to overheating of the pump, affecting its cooling efficiency and stability, and consequently impacting its service life.

Reflection Question 1.8:

Why must the vacuum pressure be below 5.0×10^{-2} Pa before turning on the quadrupole mass spectrometer?

The basic working principle of the quadrupole mass spectrometer is to use the electric field generated by the quadrupole system for mass analysis of ions. Ions are precisely controlled in the quadrupole electric field to separate ions with different mass-to-charge ratios (m/z). However, the accuracy of ion motion and detection largely depends on the vacuum environment within the system.

The quadrupole mass spectrometer generates a quadrupole electric field by controlling the direct current and radio frequency voltages on the quadrupole rods. Ions move along the axis of the quadrupole rods (typically defined as the z axis) while being constrained in the x and y directions perpendicular to the axis. For the quadrupole electric field to effectively select and stabilize or destabilize specific mass-to-charge ratios of ions, the ion motion must occur in a relatively interference-free environment. Under high vacuum conditions (below 5.0×10^{-2} Pa), the gas molecule density within the system is extremely low, significantly reducing the probability of ion collisions with background gas molecules. This environment helps to maintain the stability of ion trajectories, ensuring that they pass through the quadrupole as intended. At higher pressures, ions are more likely to collide with background gas molecules, leading to energy loss or trajectory deviation. This not only reduces the resolution of the mass spectrometer but may also result in the incorrect analysis or detection of ions with certain mass-to-charge ratios. The resolution of the quadrupole mass spectrometer depends on the stable motion of ions in the electric field. In high vacuum conditions, ion motion is least disturbed, allowing for more precise response to the selective action of the quadrupole electric field. This environment ensures that the instrument can separate ions with very close mass-to-charge ratios, achieving high-resolution mass spectrometry.

At higher pressures, background gas molecules may deposit on the surface of the quadrupole rods, forming thin layers of contaminants. These contaminants affect the uniformity of the quadrupole electric field, interfering with the ion trajectories and thereby affecting the resolution and accuracy of the mass spectrometer. Additionally, contaminants may impact the electrode characteristics of the quadrupole rods, further degrading the instrument's performance. Operating the ion source at high pressures may lead to reactions between the internal electrodes and background gas molecules, causing irreversible damage. This not only shortens the lifespan of the ion source but may also degrade the performance of the entire mass spectrometry system. Therefore, it is crucial to ensure that the system reaches the required low-pressure state before starting the ion source and quadrupole mass spectrometer to prevent such damage. At higher pressures, high voltages in the quadrupole rod system may trigger gas discharge, leading to electrical breakdown. This discharge can cause severe damage to the quadrupole system and other electronic components, potentially resulting in system failure. Maintaining a pressure below 5.0×10^{-2} Pa significantly reduces the risk of such discharges, ensuring safe operation of the equipment.

Reflection Question 1.9:

List three examples of applications in vacuum science and technology.

- Semiconductor Manufacturing

  Physical Vapor Deposition (PVD) and Chemical Vapor Deposition (CVD): These techniques are used to deposit ultra-thin layers of metals or insulating materials onto semiconductor wafers. These processes require operation in ultra-high vacuum conditions to ensure the purity and uniformity of the deposited materials and to prevent contamination by gas molecules or impurities.

  Electron Beam Lithography (E-beam Lithography): This technique is used to create nanometer-scale patterns on semiconductor wafers. The electron beam is accelerated and focused in an ultra-high vacuum environment to precisely control the etching process.

  Ion Implantation: This process involves injecting an ion beam into semiconductor materials to alter their electrical properties. The process is conducted in a vacuum environment to control the energy and depth of ion implantation, thereby achieving precise doping concentrations.

- Electron Microscopy

  Electron Beam Generation: Electron microscopes rely on high-energy electron beams generated by electron guns. This process needs to be performed in ultra-high vacuum conditions to prevent scattering of electrons due to collisions with air molecules, thus maintaining the intensity and directionality of the electron beam.

  Sample Chamber Vacuum: The sample chamber is also maintained at high vacuum to reduce interference from air molecules on the electron beam and to ensure the sample surface remains uncontaminated. This allows the electron microscope to obtain clear images of samples at the nanometer or even atomic level.

  Scanning and Transmission Imaging: In Scanning Electron Microscopy (SEM) and Transmission Electron Microscopy (TEM), the electron beam scans or passes through the sample. The vacuum environment ensures the stability of the electron beam and the cleanliness of the sample surface, enhancing image resolution and contrast.

- Space Simulation and Aerospace Testing

  Space Environment Simulation: During the design and testing of spacecraft and satellites, it is necessary to simulate the high-vacuum conditions of outer space. Vacuum chambers are used to replicate the low-pressure conditions, high radiation environment, and extreme temperature variations of space to test the performance and reliability of spacecraft and materials under real space conditions.

  Spacecraft Component Testing: Components such as solar panels, thermal control systems, and propulsion systems need to be tested in high-vacuum environments to ensure their proper functioning in space. Space simulation includes not only low pressure but also high radiation and temperature extremes, allowing for precise reproduction of space conditions on the ground.

  Spacesuit Testing: Vacuum chambers are also used to test the airtightness, thermal insulation, and pressure resistance of spacesuits to ensure astronaut safety during spacewalks.

Reflection Question 1.10:

List three examples of applications in plasma science and technology.

- Plasma Processing

  Plasma Etching: In semiconductor manufacturing, plasma etching technology is used to etch fine patterns of microelectronic devices. By controlling the chemical composition and energy of the plasma, precise etching of silicon wafers or other materials can be achieved to form the desired circuit structures. Plasma etching offers high precision, high anisotropy, and low damage, and is widely used in integrated circuit fabrication.

  Plasma Surface Modification: Plasma technology can be used to alter the physical and chemical properties of material surfaces, such as enhancing surface energy through plasma treatment to improve adhesion, wettability, or biocompatibility. This technology is widely applied in fields such as medical devices, packaging materials, and automotive components.

  Plasma Spraying: In the processing of metals and ceramics, plasma spraying is used to deposit high-performance coatings on substrates to improve wear resistance, corrosion resistance, and high-temperature resistance. This technology has significant applications in aerospace, energy, and automotive industries.

- Plasma Display Panel (PDP)

  Plasma displays generate images by applying voltage within tiny gas chambers to excite gas discharge and produce ultraviolet light, which then excites phosphors to emit visible light. Each pixel contains sub-pixels of red, green, and blue, and the brightness of these sub-pixels is controlled by plasma discharge to produce full-color images. Plasma displays are characterized by high contrast and wide viewing angles, making them suitable for large-size TVs and public displays. Although the market share of PDPs has declined with the development of LCD and OLED technologies, plasma displays still offer unique advantages in specific applications, such as in scenarios requiring high brightness and fast response. Each pixel in a plasma display can emit light independently and respond quickly, enabling high refresh rates and fast response times, making it particularly suitable for displaying fast-moving images, such as sports events or action movies.

- Nuclear Fusion Research

  Magnetic Confinement Fusion: In nuclear fusion experiments, such as those conducted with Tokamak devices, plasma is heated to extremely high temperatures (tens of millions to hundreds of millions of degrees Celsius) to enable fusion reactions of hydrogen isotopes (e.g., deuterium and tritium). Magnetic fields are used to confine the plasma and prevent it from contacting the container walls, thereby maintaining the high-temperature and high-density conditions necessary for controlled nuclear fusion.

  Inertial Confinement Fusion (ICF): In inertial confinement fusion experiments, powerful lasers or particle beams are focused on a small fuel pellet, creating intense plasma shock waves that compress the fuel to extremely high densities and temperatures, triggering nuclear fusion reactions. The formation and behavior of the plasma are crucial for energy transfer and fusion reaction efficiency.

  Plasma Diagnostics: In nuclear fusion research, accurate diagnostics of plasma parameters such as temperature, density, and velocity distribution are essential for understanding and controlling fusion reactions. Plasma diagnostic techniques include spectroscopy, interferometry, and Thomson scattering, which enable real-time monitoring of plasma conditions and optimization of reaction parameters.

Major:	Physics	Grade:	2022
Name:	杨舒云	Student number:	22344020
Room temperature:	20°C	Experimental location:	A101
Student Signature:	**In Attachment**	Score:	
Experiment time:	2024/12/20	Teacher's Signature:	

APL1-2 Vacuum Compatibility Testing of Materials and Plasma Characteristics Study Experimental Record

2.1 Content, Procedures & Results

2.1.1 Operations

1. 使用机械泵和分子泵获得高真空。

(1) 启动机械泵观察记录真空度（真空计压强）随时间（5 分钟）的变化。机械泵先抽真空压强低于 10 Pa 后，启动分子泵观察记录真空度（真空计压强）随时间的变化，待分子泵达到额定转速（约需 8 分钟）后再观察记录（5 分钟）。

(2) 停止分子泵观察记录真空度（真空计压强）随时间的变化（分子泵转速降为零约需 8 分钟）。待分子泵完全停止后，关闭机械泵，记录真空度（真空计压强）随时间的变化（5 分钟）。

实验结果如**Figure 3**和**Figure 4**所示。

2. 通过真空气体放电实验，测量击穿电压与电极间隙和气压之间的关系，验证帕邢定律，了解气体放电基本物理过程。

(1) 检查实验设备：

- 测量电极间实际间距。
- 确保放电管与电源连接可靠，电压和气体流量旋钮均调至最小位置。
- 确保高压电源开关和分子泵电源开关处于断开状态。

(2) 打开总电源，启动机械泵，抽真空至 2-3 Pa（约 10 分钟）。

(3) 调节减压阀，使流量计前气压保持在 0-1 个大气压（由指导教师操作）。

(4) 调节流量计，使放电管内气压稳定在 20 Pa，记录真空计读数。

(5) 打开高压电源开关，逐步升高电压：

- 快速升至 200 V，再缓慢升高直至气体击穿，记录击穿电压及对应气压。
- 缓慢降压至 0 V，观察放电熄灭电压，并注意与击穿电压的差异。

(6) 调整气压并重复测量：

- 增加气体流量，每次升高气压至 30 Pa、40 Pa、...、100 Pa，重复击穿电压测量（共 8 组数据）。
- 减小气压至 20 Pa，重复击穿测量。
- 逐步减小气压，每次减少约 2 Pa，直至 4 Pa，重复测量（共 7-8 组数据）。

(7) 实验结束：

- 将气体流量和电压调至最小值。
- 依次关闭高压电源、机械泵、电源总开关。

实验结果如**Figure 5**所示。

3. 观察气体放电（发光）现象，利用光纤光谱仪研究气体放电产生的等离子体光谱特性，获得等离子体基本参数，了解等离子体物理的基本知识。

(1) 准备实验装置：

- 检查光纤光谱仪、放电管及电源连接是否可靠。
- 确保气体流量和电压调节旋钮处于最小位置。

(2) 启动真空系统，调节气压至所需值（如 20 Pa）。

(3) 打开高压电源，缓慢增加电压，观察气体放电产生的发光现象。

(4) 使用光纤光谱仪记录不同气压和电压条件下的等离子体光谱。

(5) 分析光谱数据，提取等离子体的基本参数（如电子温度、离子密度）。

(6) 实验结束：

- 将气体流量和电压调至最小值。
- 依次关闭高压电源、真空泵、电源总开关。

实验结果如如**Figure 6**、**Figure 7**和**Figure 8**所示。

4. 了解四极质谱仪工作原理，使用四极质谱仪进行真空系统检漏和气体成分分析。

(1) 检查实验装置：

- 确保真空腔体、电源、四极质谱仪的电路及数据线连接可靠。
- 电压调节旋钮和气体流量调节旋钮均置于最小位置。
- 确保高压电源、分子泵和应急开关均处于断开状态。
- 检查真空腔体的气密性。

(2) 打开总电源，启动机械泵，抽真空至 10 Pa 以下（约需 10 分钟）。

(3) 在机械泵压强达到 10 Pa 以下时，开启分子泵：

- 启动分子泵电源，按下绿色按钮，等待自检完成后，转速从 68 rpm 上升至 450 rpm。

(4) 当真空计显示压强低于 5.0×10^{-2} Pa，打开四极质谱仪。

(5) 向真空腔内加入适量 He、Ne、Ar 等气体（压强不得超过 5.0×10^{-2} Pa），通过软件观察气体的质谱图。

(6) 打开软件 `VAccuRay3.0`：

- 右键选择 `Device Setup`，保持默认参数，点击 `OK`。
- 点击灯丝选项，确认真空度低于 5.0×10^{-2} Pa 后启动灯丝。
- 观察并记录质谱图上的气体波峰。

(7) 修改实验参数（如 `Resolution`, `Emission`, `eVolt` 等），重复测量并记录质谱数据。

(8) 实验结束：

- 按下分子泵电源面板的红色按钮，待转速降至 0 后关闭分子泵电源，再关闭机械泵。
- 最后关闭总电源，整理数据并总结实验结果。

实验结果如**Figure 9**、**Figure 10**、**Figure 11**和**Figure 12**所示。

2.1.2 Display

1.【实验一】

利用实验测定的数据绘图，分别得到抽真空过程压强-时间关系如图**Figure 3**所示，回升过程压强-时间关系如**Figure 4**所示。

测量得到的极限真空值为：7.5×10^{-3}Pa。

2.【实验二】

首先，两电极间距固定，为 $d = 60mm$。

利用实验测定的数据绘图，如**Figure 5**所示。可以发现，击穿电压有最低值，且数据分布基本符合 $y = \frac{x}{\ln x}$ 形式。

3.【实验三】

通过实验，测定 He、Ar 和 Ne 的放电光谱，然后编写脚本寻找了三条主要谱线，结果如**Figure 6**、**Figure 7**和**Figure 8**所示。

4.【实验四】

对上述实验结果和数据的分析与讨论详见数据分析部分。

2.2 Original Data

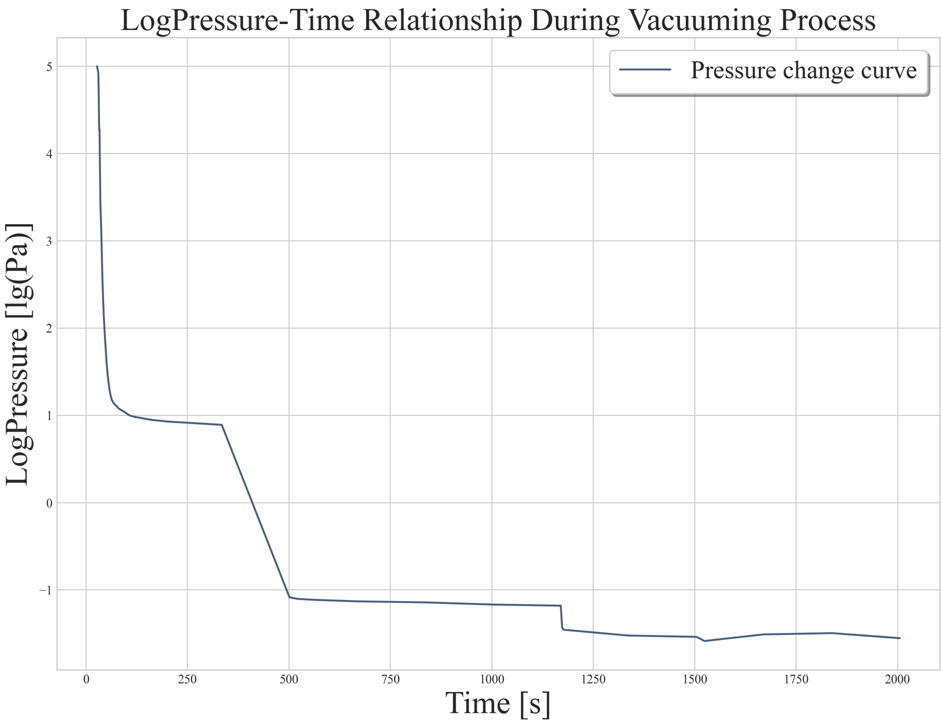

Figure 3: 抽真空过程压强-时间关系

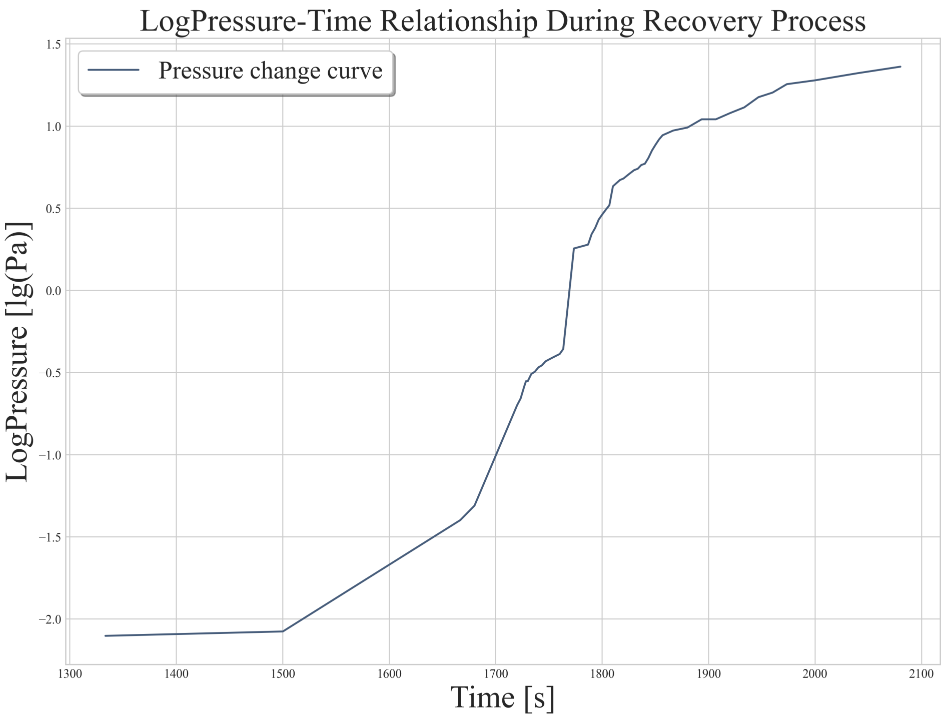

Figure 4: 回升过程压强-时间关系

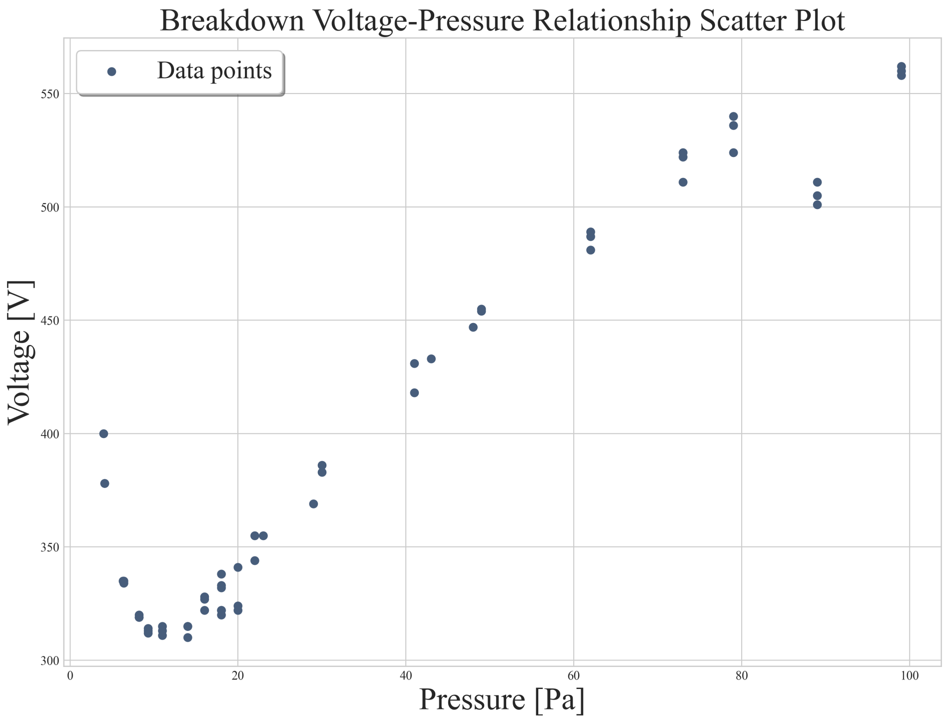

Figure 5: 击穿电压-压强关系（实验测定数据点）

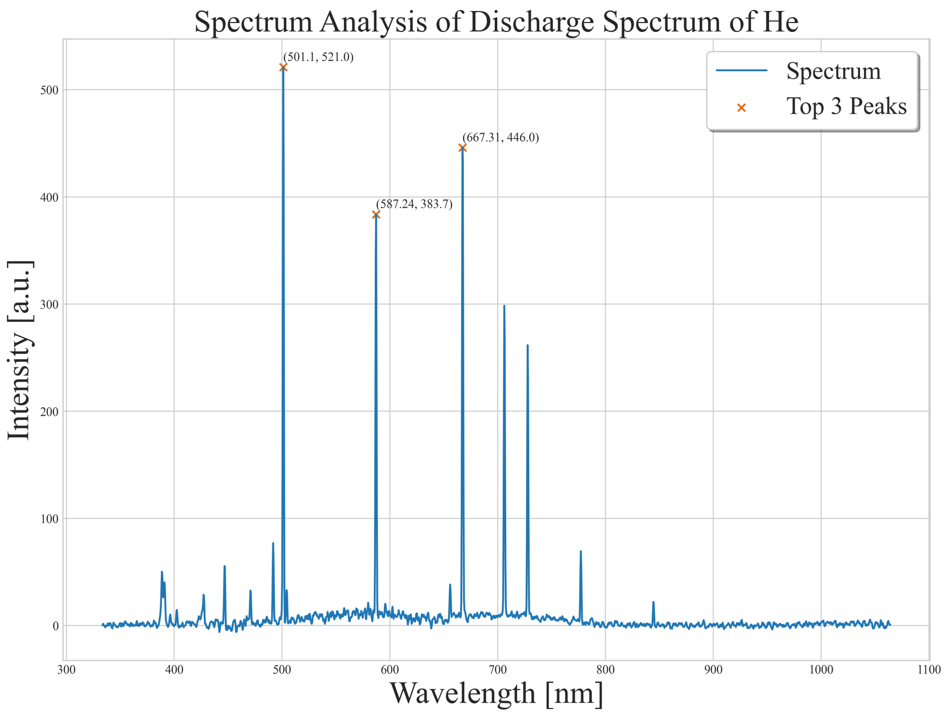

Figure 6: He 放电谱线

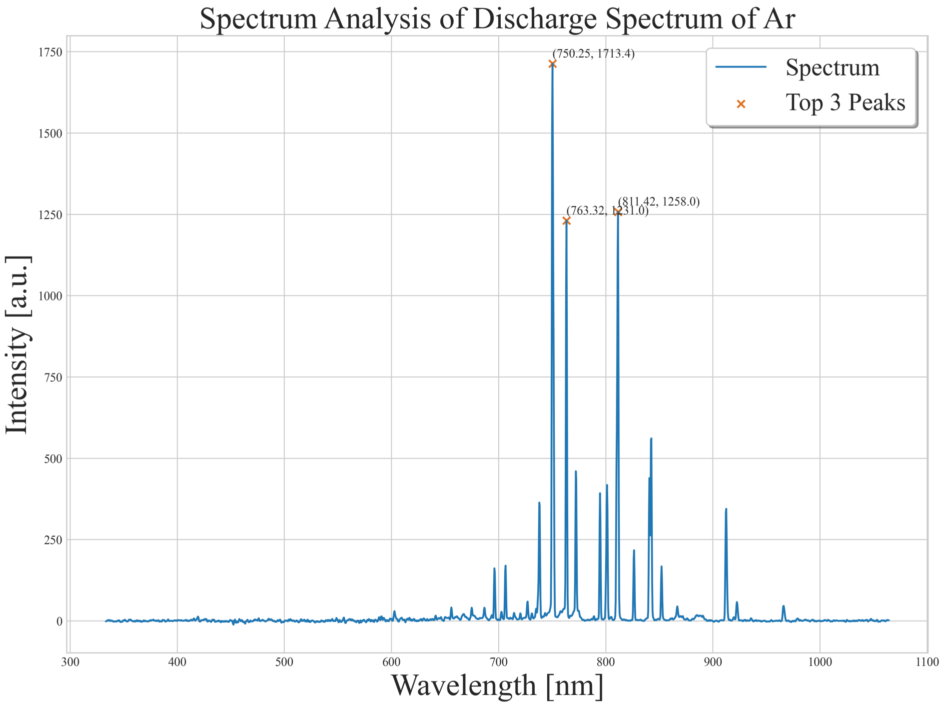

Figure 7: Ar 放电谱线

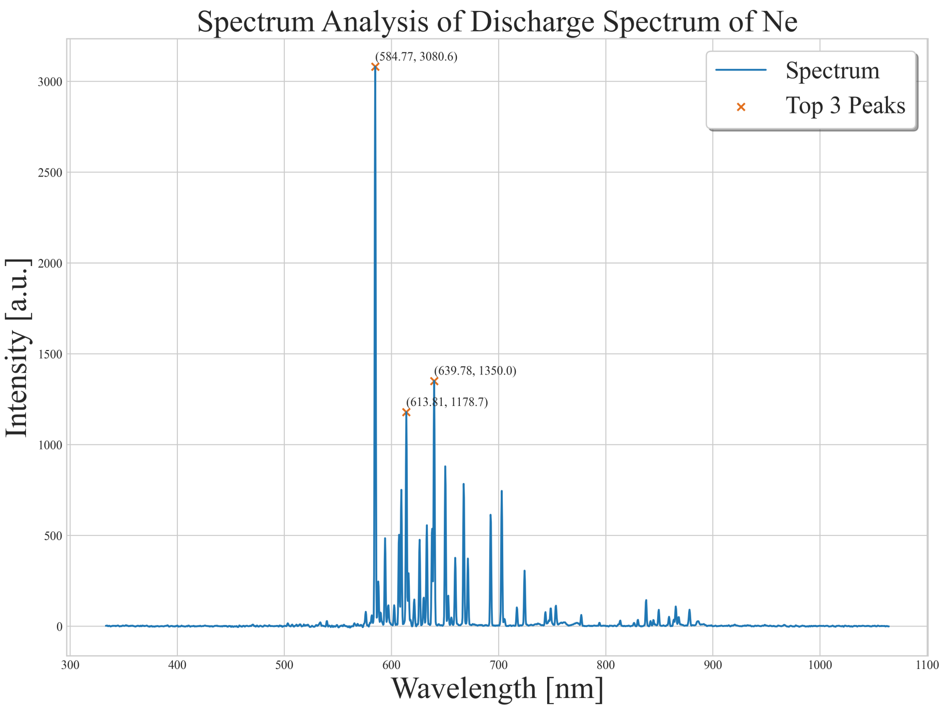

Figure 8: Ne 放电谱线

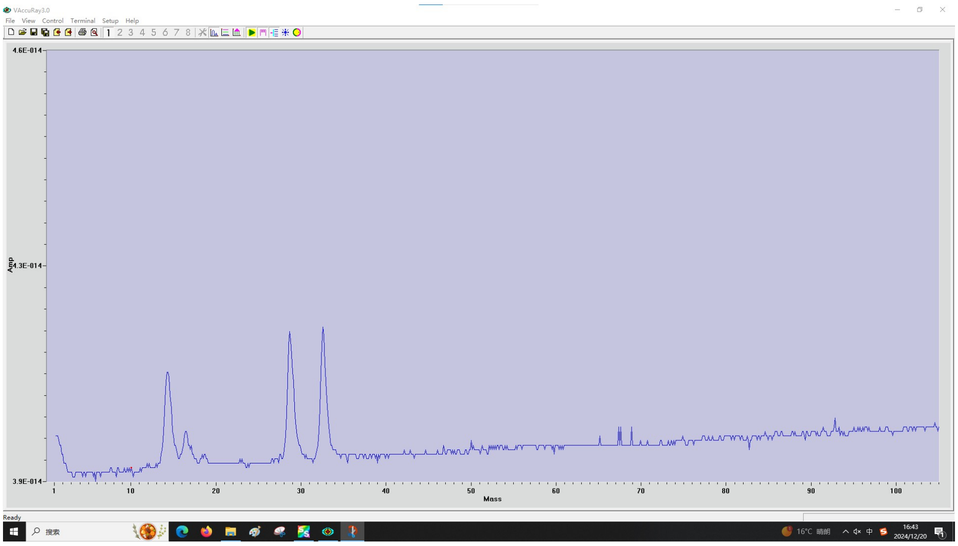

Figure 9: 空气的质谱仪谱线（实验测定）

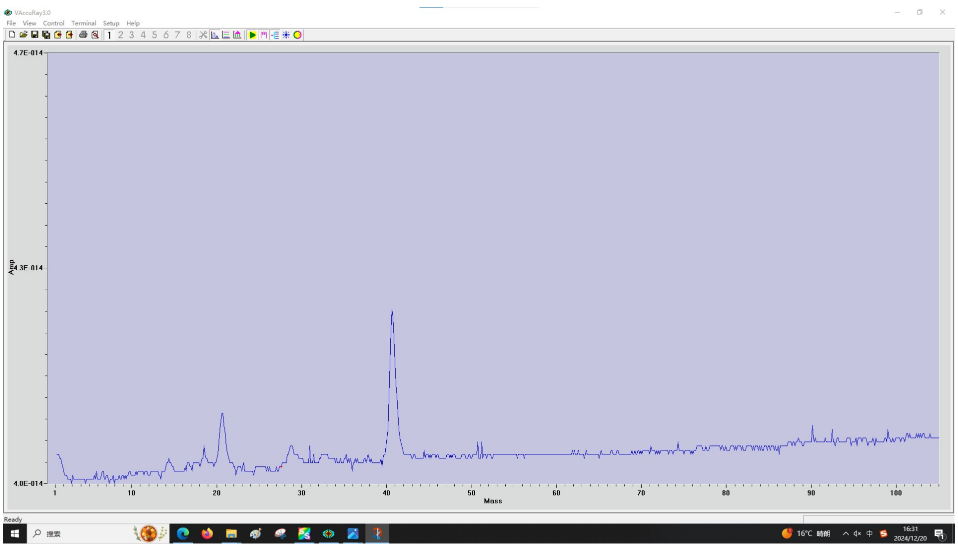

Figure 10: Ar 的质谱仪谱线（实验测定）

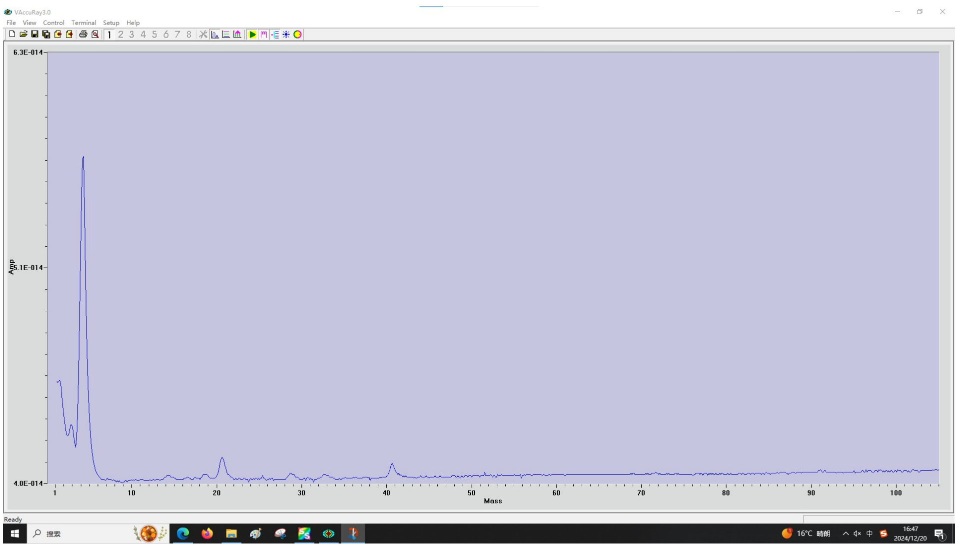

Figure 11: He 的质谱仪谱线（实验测定）

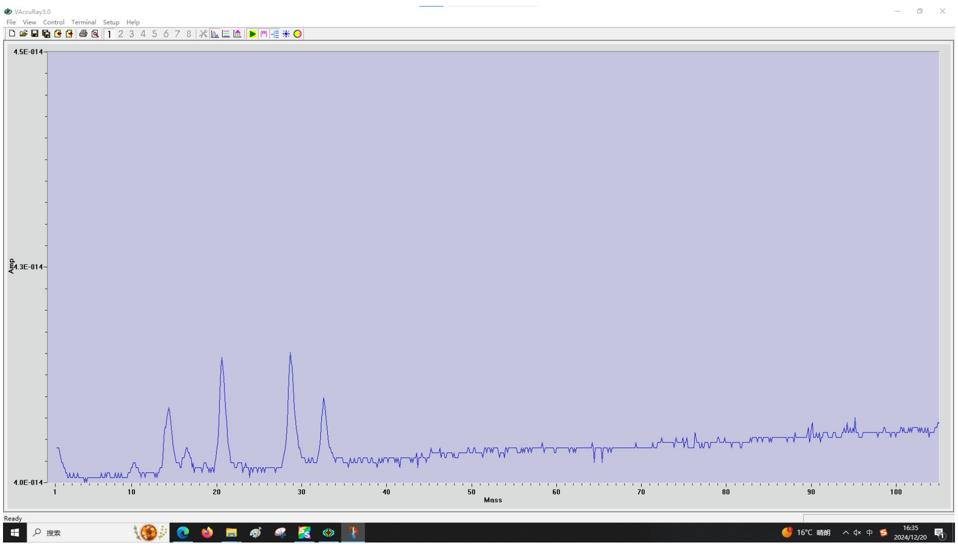

Figure 12: Ne 的质谱仪谱线

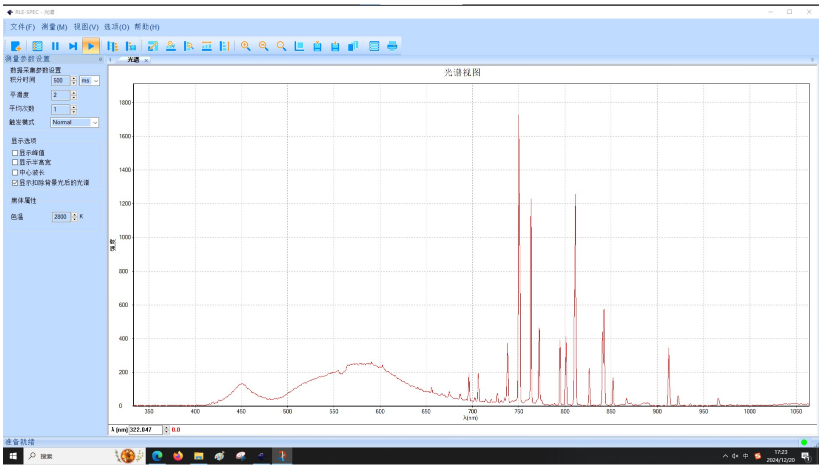

Figure 13: 原始数据展示

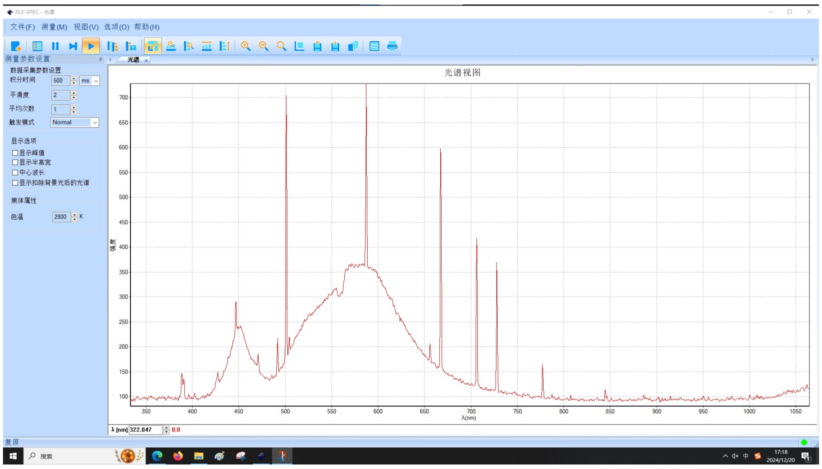

Figure 14: 原始数据展示

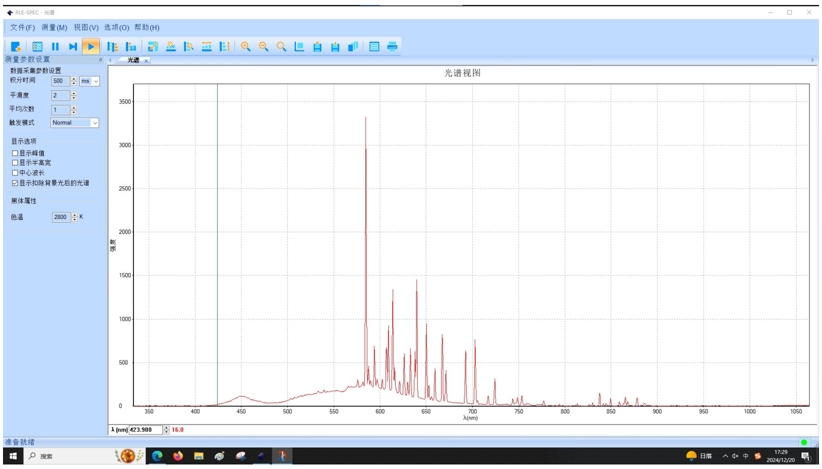

Figure 15: 原始数据展示

Major:	Physics	Grade:	2022
Name:	杨舒云	Student number:	22344020
Date:	2024/12/30	Score:	

APL1-2 Vacuum Compatibility Testing of Materials and Plasma Characteristics Study
Analysis & Discussion

3.1 Data Processing

3.1.1 Analysis

1. 通过真空气体放电实验，测量击穿电压与电极间隙和气压之间的关系，验证帕邢定律，了解气体放电基本物理过程。

   根据 Paschen 定律，考虑采用函数 $V_d = \frac{Ap}{\ln(Bp)}$ 进行拟合，拟合结果如**Figure 18**所示。

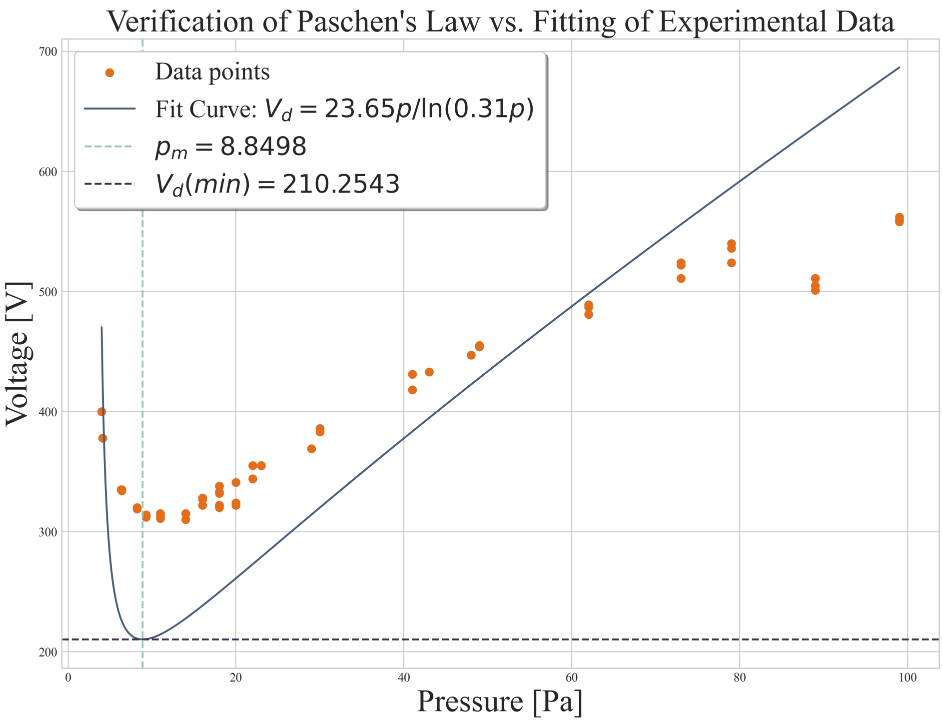

Figure 18: 击穿电压-压强关系（拟合）

   从拟合结果看出，实验结果基本验证了 Paschen 定律。

2. 测定了 Ar 气体放电产生的等离子体光谱后，可以通过以下方法分别求解电子温度 (T_e)、电子密度 (n_e)

和离子密度 (n_i)。

电子温度 (T_e) 反映了等离子体中自由电子的平均能量，可以通过谱线强度比法 (Boltzmann Plot Method) 或两谱线强度比法求解。在 Boltzmann 图法中，利用公式：

$$\ln\left(\frac{I_{ij}\lambda_{ij}}{g_i A_{ij}}\right) = -\frac{E_i}{k_B T_e} + \ln\left(\frac{hcN}{Z(T)}\right)$$

其中，I_{ij} 是谱线强度，λ_{ij} 是谱线波长，g_i 是能级的简并度，A_{ij} 是自发辐射跃迁几率，E_i 是上能级能量，$Z(T)$ 是配分函数，k_B 是玻尔兹曼常数，h 是普朗克常数，c 是光速。选择多个高能跃迁谱线，记录其强度和波长，查表获取对应的 $\{E_i, g_i, A_{ij}\}$，绘制 $\ln(I_{ij}\lambda_{ij}/g_i A_{ij})$ 对 E_i 的关系图，斜率为 $-1/(k_B T_e)$。或通过两谱线强度比公式：

$$\frac{I_1}{I_2} = \frac{g_1 A_1}{g_2 A_2}\exp\left(-\frac{E_1 - E_2}{k_B T_e}\right)$$

选取两条已知能级差的谱线，通过强度比直接求解 T_e。

电子密度 (n_e) 通常通过谱线加宽效应，特别是斯塔克加宽来测定。斯塔克加宽公式为：

$$\Delta\lambda_{Stark} = 2w \cdot n_e^{2/3}$$

其中，$\Delta\lambda_{Stark}$ 是谱线的斯塔克加宽（半高全宽，FWHM），w 是与谱线跃迁相关的斯塔克加宽参数。选择无显著自吸效应的谱线，测量其半高全宽 $\Delta\lambda_{Stark}$，结合查表所得的 w 值，通过公式反推出 n_e。若存在显著自由-自由或自由-束缚辐射，也可利用背景辐射强度与电子密度的关系推导 n_e。

离子密度 (n_i) 则通过电离平衡条件结合电子温度和电子密度推算，常用萨哈方程：

$$\frac{n_{i+1}n_e}{n_i} = \frac{2Z_{i+1}}{Z_i}\left(\frac{2\pi m_e k_B T_e}{h^2}\right)^{3/2}\exp\left(-\frac{E_{ion}}{k_B T_e}\right)$$

其中，n_i, n_{i+1} 是离子化前后离子的密度，Z_i, Z_{i+1} 是配分函数，E_{ion} 是电离能，m_e 是电子质量。结合光谱中离子谱线强度和已知的 T_e、n_e，可迭代计算 n_i。此外，还可以通过谱线相对强度结合配分函数和电离能参数推导离子密度。

然而，由于实验室 Ar 与 Ne 均不纯，He 相对纯净，以下以 He 给出上述分析的示例。

(1) 光谱测量数据：

- 主峰位置：501.099 nm
- 主峰强度：521.0 arb. units
- 半高全宽 (FWHM)：1.1545 nm

(2) 电子温度 (T_e)：使用 Boltzmann 图法，选取氦光谱线 (587.244 nm, 667.311 nm, 501.099 nm) 的能级信息，计算得到：

$$T_e = 0.94\,\text{eV}$$

(3) 电子密度 (n_e)：基于主峰 (501.099 nm) 的半高全宽 (1.1545 nm) 和斯塔克加宽参数 ($w = 0.05\,\text{nm}\cdot\text{cm}^3$)，计算得到：

$$n_e = 2.71\times 10^{19}\,\text{cm}^{-3}$$

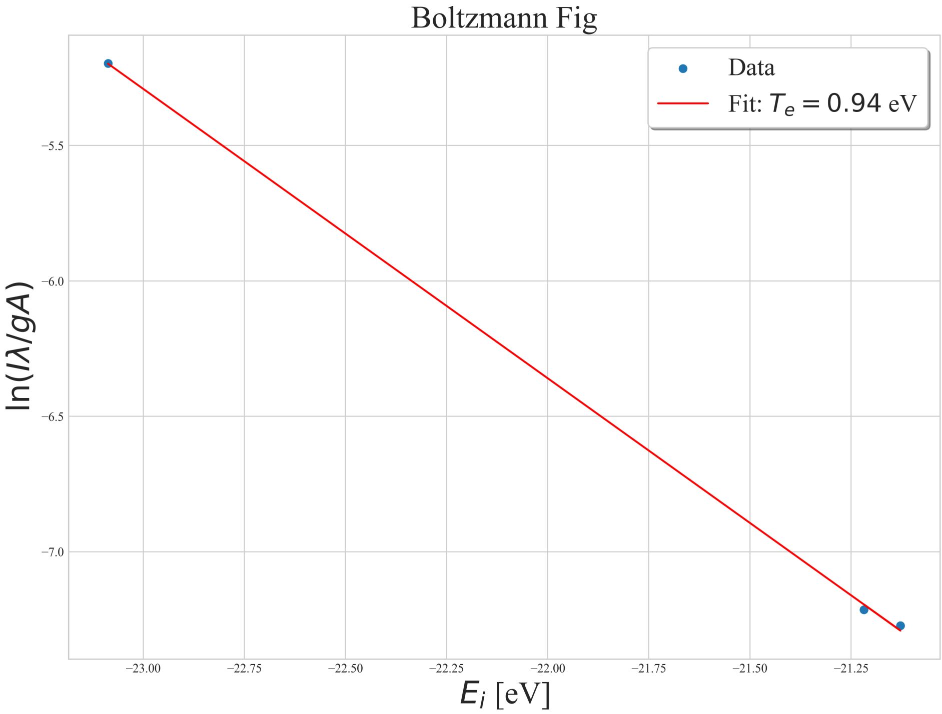

Figure 19: He-Boltzmann 图

(4) **离子密度** (n_i)：假设总粒子密度 $n_{\text{total}} = 1 \times 10^{19}\,\text{cm}^{-3}$，基于萨哈方程计算得：

$$n_i \approx 9.99 \times 10^{18}\,\text{cm}^{-3}, \quad n_{i+1} \approx 3.21 \times 10^{13}\,\text{cm}^{-3}$$

3. 质谱仪谱线分析，考虑扣除如**Figure 20**所示本底。

然后利用寻峰算法寻找主要谱线，结果见讨论部分。

3.1.2 Discussion

1. 基于 He 放电光谱讨论其等离子体参数特性

(1) 测定了氦气放电等离子体的关键参数，包括电子温度（$T_e = 0.94\,\text{eV}$）和电子密度（$n_e = 2.71 \times 10^{19}\,\text{cm}^{-3}$），表明该等离子体具有典型的低温高密度特性。离子密度（n_i）显著高于一次电离态（n_{i+1}）的密度，符合氦气高电离能的特性。

(2) 等离子体主要由中性粒子和少量一次电离的离子组成，电子与离子的相互作用主导了其光谱特性。电子密度较高，表明等离子体中存在显著的碰撞过程。

(3) Boltzmann 图法和斯塔克加宽法提供了可靠的参数估算手段，结合萨哈方程进一步拓展了对离子分布的理解。

2. 质谱仪测定结果讨论

(1) 空气

- 质量数 14.4 和 28.7：质量数约为 14.4 的峰可能对应于 N+（氮原子离子）。质量数约为 28.7 的峰可能对应于 N2+（氮气分子离子），氮气占空气的主要组成部分（ 78%）。

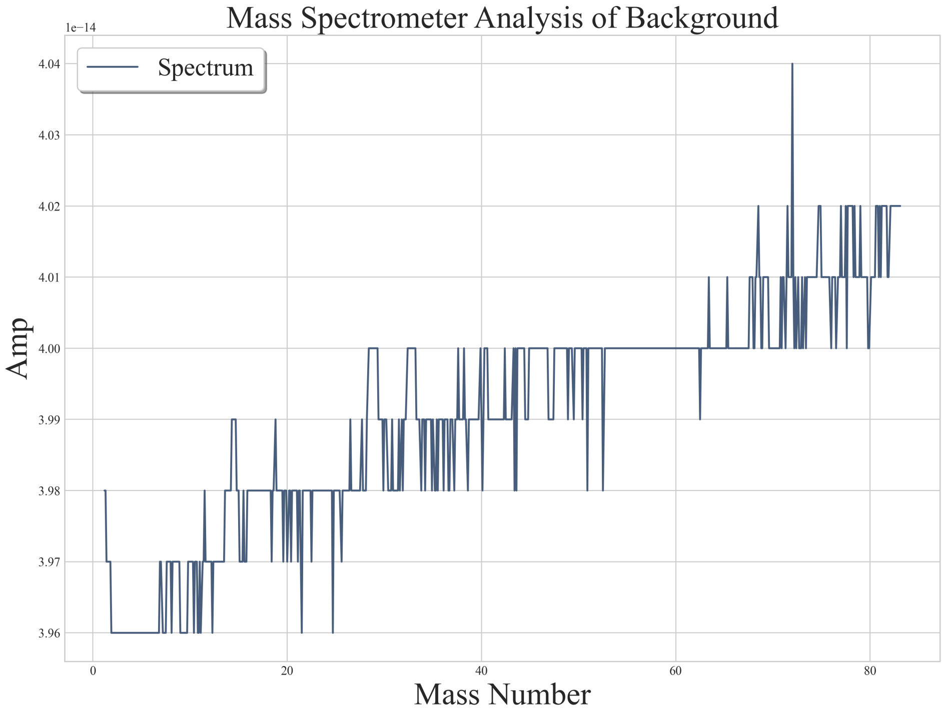

Figure 20: 质谱仪背景

- 质量数 32.6：质量数约为 32.6 的峰可能对应于 O2+（氧气分子离子）。氧气是空气的第二大成分（ 21%）。
- 其他小峰：小峰可能对应于少量的稀有气体（如氩 Ar）或其他痕量成分（如 CO2、H2O）。

(2) Ar

- 质量数 40.7：这是 Ar 的主要同位素 40Ar 的离子峰。氩气的主要同位素是 40Ar，占比约 99.6%，因此这是谱图中最显著的峰。
- 质量数 20.6 和 18.5：这些可能是仪器背景噪声或少量氩气中的其他同位素贡献，如 36Ar 或 38Ar，但信号偏低可能也与质谱仪的检测灵敏度和分辨率有关。
  20.6 还有一种更有可能的结果：40Ar 的二价离子（电荷数为 2）。
  18.5 的峰可能是水分子（H2O）的背景信号，因为实验环境中的水蒸气常见于质谱背景。

(3) He

- 质量数 4.2：这是 He+（氦原子离子）的峰，对应氦气的主要同位素 4He。氦气几乎完全由 4He 组成（约 99.999%），因此该峰是谱图中最显著的。
- 质量数 2.9：该峰可能是背景信号或氢分子离子 H2+ 的干扰。氢分子（H2）容易存在于实验室环境或真空系统中，可能导致这一干扰峰。
- 质量数 40.7：该峰与之前分析的氩气谱图一致，可能是 Ar+（氩气离子）的背景信号。即使实验室环境中仅有微量的空气或残余氩气，也会在高灵敏度下被检测到。
- 谱图中明显只有单一主峰，表明氦气纯度较高。

(4) Ne

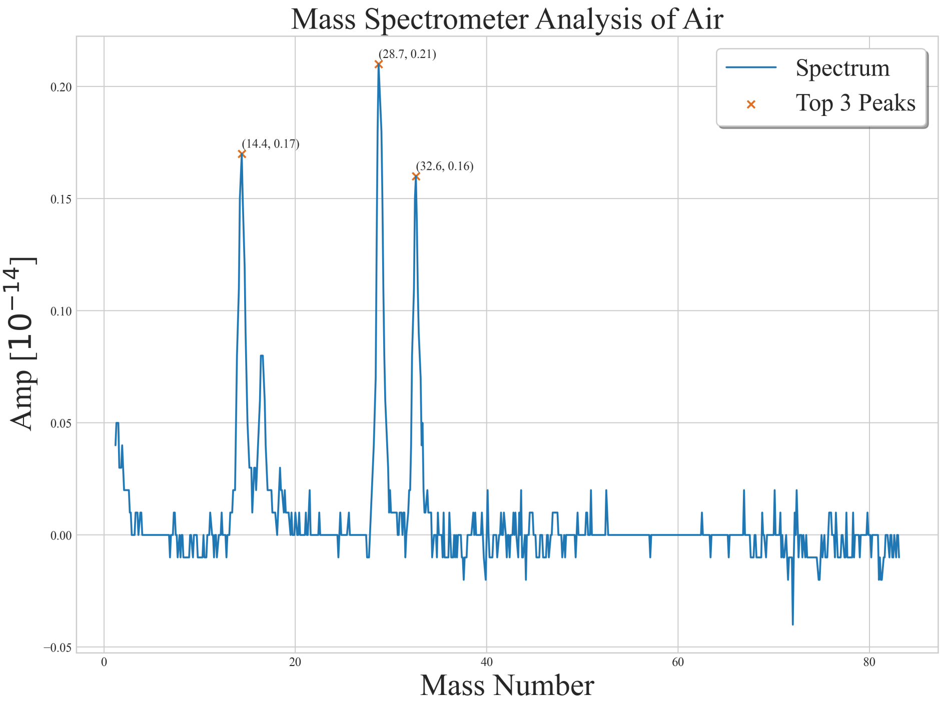

Figure 21: 空气的质谱仪谱线分析

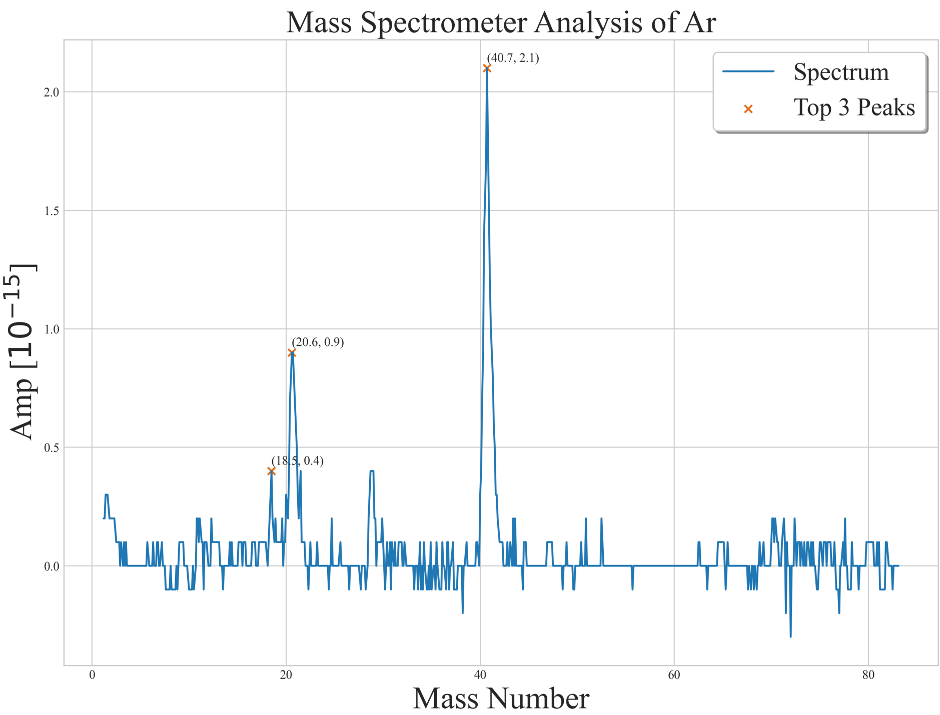

Figure 22: Ar 的质谱仪谱线分析

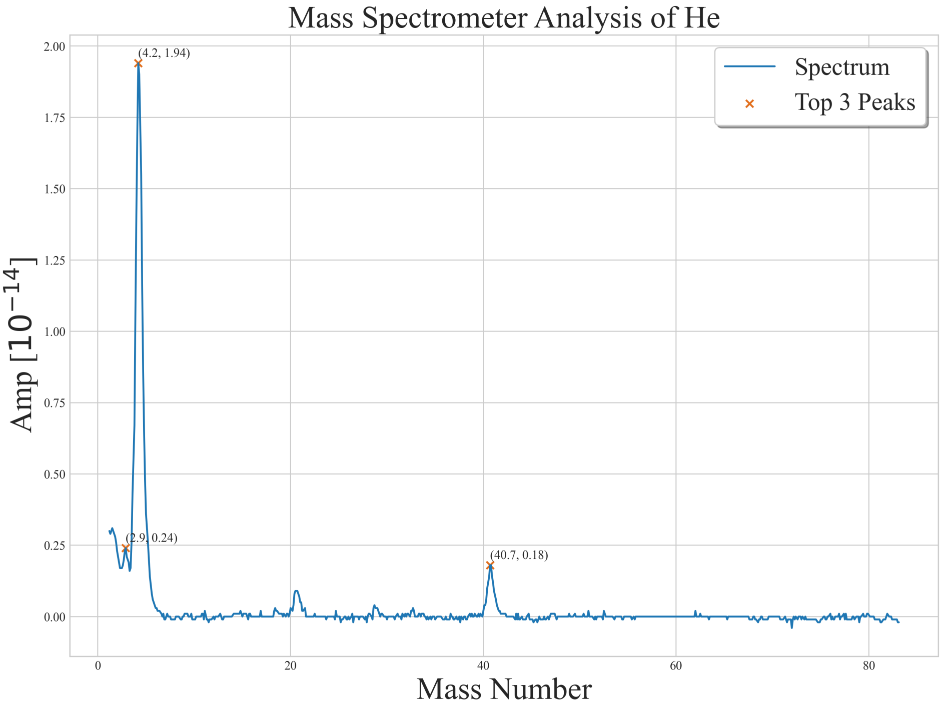

Figure 23: He 的质谱仪谱线分析

- 质量数 20.6（主要峰）：这是 20Ne 的离子峰，对应氖气的主要同位素，约占 90.48%。该峰是谱图中最显著的，符合高纯度氖气的主要特征。
- 质量数 22.7（次峰，未标出）：可能是 22Ne 的离子峰，22Ne 是氖气的次要同位素，约占 9.25%。强度通常较主峰（20Ne）小，但应能分辨。
- 质量数 14.3 和 28.7：14.3：可能是 N+（氮离子），说明实验环境中可能有少量残余氮气。28.7：可能是 N2+（氮气分子离子）或背景中空气的贡献。
  谱图中噪声较多，可能与真空系统中的其他气体污染有关，如空气中的氧气、氩气等。

3.2 Reflections after Experiment

Reflection Question 3.1:

真空度（真空计压强）随时间变化反映了真空系统的什么特性?

真空度（真空计压强）随时间的变化能够反映真空系统的多个关键特性，是评估真空性能的重要指标之一。首先，真空度的变化直接体现了真空泵的抽气性能。初始阶段，真空度下降的速度越快，说明真空泵的抽气速率较高，系统排气效率较好。而在接近极限真空时，真空度下降变缓，反映了真空泵在高真空状态下的极限能力和稳定性。这一过程有助于判断真空泵的性能是否符合系统需求。

随着时间推移，真空度达到一定值后可能出现缓慢回升或波动，这通常是系统漏气或材料放气的表现。如果真空度呈线性回升，可能是密封件或法兰接口存在漏气问题；如果真空度在某个范围内波动，则可能是腔体壁或材料表面吸附的气体逐渐释放。材料的放气特性在真空度变化中也起到了重要作用，尤其是在抽气初期，吸附气体的解吸过程会减缓真空度的下降速度。当放气速率与抽气速率趋于平衡时，系统真空度会稳

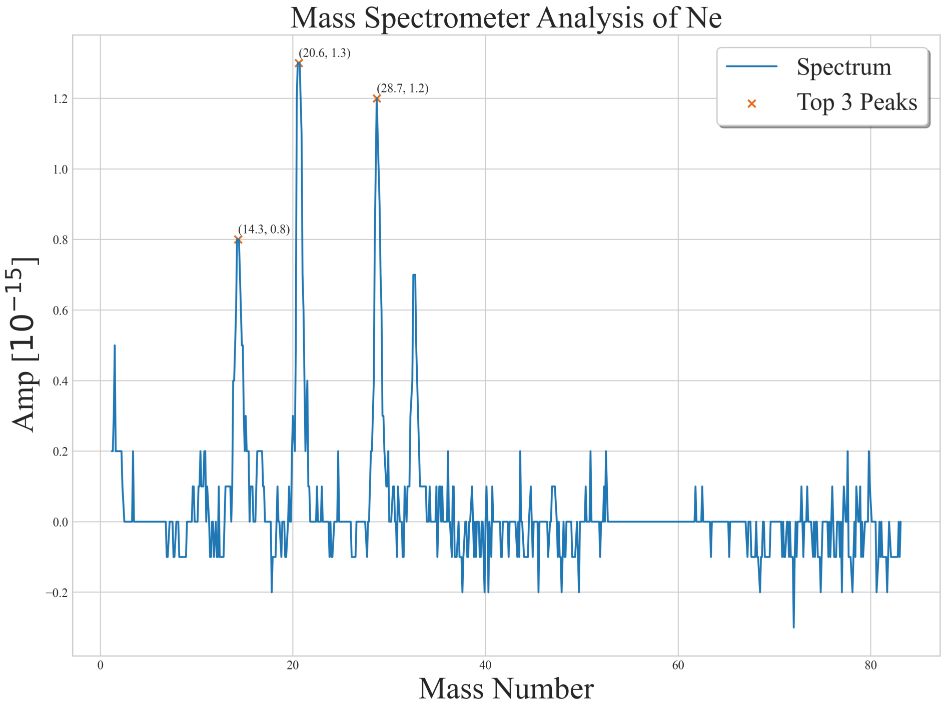

Figure 24: Ne 的质谱仪谱线分析

定在某一值，这时可以评估系统材料的洁净度和气体释放速率。

真空系统的极限真空，即最低达到的压强，体现了系统的综合性能，包括真空泵的能力、密封件的可靠性以及材料的洁净度。如果极限真空较低，说明系统的整体真空性能较优；反之，则可能需要检查密封件或腔体清洁程度。此外，在气体动态引入或调整真空系统参数时，真空度随时间的响应速度可以反映系统的动态特性。响应速度较快的系统通常表明气体流路设计合理；而响应较慢则可能受到管道阻力或死体积效应的影响。

通过记录真空度随时间的变化并进行分析，可以全面了解真空系统的性能。这不仅能够评估真空泵和系统的稳定性，还能诊断可能存在的漏气或放气问题，确保系统在实验前达到最佳状态。同时，这些数据还可以用于预测真空系统达到稳定状态所需的时间，为复杂实验的准备和运行提供可靠依据。

详见原理概述部分。

Reflection Question 3.2:

等离子体光谱反映了等离子体的什么特性？能得到等离子体的什么参数？

等离子体光谱反映了等离子体的多种物理特性，是研究等离子体状态的重要工具。首先，光谱直接反映了等离子体的成分特性。通过光谱线的波长，可以识别等离子体中存在的原子、离子及分子种类，从而确定其化学组成。同时，不同谱线的强度和位置反映了粒子的激发状态和能级分布，揭示了等离子体的能量传输特性。此外，光谱的分布还可以判断等离子体是否处于局部热力学平衡 (LTE)，即粒子的激发和电离是否遵循玻尔兹曼分布和萨哈平衡。

利用等离子体光谱，可以获得多种关键参数。首先，通过分析谱线强度比，特别是不同能级跃迁的强度比，可以计算电子温度 (T_e)，这一参数描述了自由电子的平均能量，是等离子体的重要热力学特性。其次，

通过谱线的斯塔克加宽，可以推算电子密度 (n_e)，这一参数反映了等离子体中自由电子的浓度。此外，通过综合光谱数据，结合电离平衡条件，可以进一步推算离子密度 (n_i) 和粒子间的相对丰度。

光谱的形状和宽度还可以提供有关等离子体动力学特性的信息。例如，谱线的多普勒加宽反映了粒子的热运动速度，而斯塔克加宽则与局部的电场强度和电子密度相关。结合这些信息，可以全面了解等离子体的局部微观状态，包括粒子的碰撞和能量交换机制。此外，通过分析连续光谱（自由-自由辐射和自由-束缚辐射），可以提取等离子体的光学厚度和辐射特性，为研究能量输运和电磁波与等离子体相互作用提供基础。

详见原理概述部分。

Reflection Question 3.3:

研究四极电场特性及其中离子运动方程，深入探讨四极质谱仪工作原理。

四极电场由四根平行排列的金属电极组成，这些电极形成了一个对称的正方形结构。通过施加交流电压和直流电压的组合，可以在空间中产生特定分布的电场，其电势分布为 $\Phi(x,y,t)=\frac{U+V\cos(\omega t)}{r_0^2}(x^2-y^2)$，其中 U 是直流电压，V 是交流电压的幅值，$\omega=2\pi f$ 是交流电压的角频率，r_0 是电极中心到表面的距离。这一分布产生的电场仅在 x 和 y 平面内具有分量，对离子运动施加周期性的力，而在 z-方向没有电场分量。

离子在四极电场中的运动遵循牛顿第二定律，其在 x 和 y 方向的运动方程分别为 $\frac{d^2x}{dt^2}+\frac{q}{m}\frac{U+V\cos(\omega t)}{r_0^2}x=0$ 和 $\frac{d^2y}{dt^2}-\frac{q}{m}\frac{U+V\cos(\omega t)}{r_0^2}y=0$。这些方程为标准的 Mathieu 方程，解的稳定性决定了离子是否能够在四极电场中被传输或偏转。Mathieu 方程的一般形式为 $\frac{d^2u}{d\xi^2}+[a-2q\cos(2\xi)]\,u=0$，其中参数 a 和 q 的具体表达式为 $a_x=\frac{4qU}{m\omega^2r_0^2}$，$q_x=\frac{2qV}{m\omega^2r_0^2}$，而在 y 方向的对应值为负。通过分析 Mathieu 方程的稳定性解，可以确定离子在 x 和 y 方向的稳定区域。只有在两个方向均稳定的离子才能通过四极电场。

基于上述原理，四极质谱仪通过调整 U 和 V 的值来筛选离子质量。当 U/V 比固定时，离子的质量与电场的稳定区域直接相关，只有满足稳定性条件的离子才能通过四极电场，其它离子则会因运动不稳定而撞击电极，无法到达检测器。通过调节 U 和 V 的幅值，质谱仪可以扫描不同质量的离子。在定频模式下，保持交流频率 ω 不变，通过调整 U 和 V 的比例选择目标质量离子。而在扫描模式下，逐步改变电压参数，使得不同质量的离子依次进入稳定区域并被检测。

离子在四极电场中的稳定性不仅取决于电场参数，还受电极几何和离子初始条件的影响。分辨率由电压调控精度、电场对称性和离子稳定区域宽度决定。灵敏度则与检测器效率、电场稳定性和离子信号强度密切相关。高分辨率要求精确的电压控制，而高灵敏度则依赖于优化的信号检测和电场设计。

详见原理概述部分。